

Refractive Gravity: An Ontological Picture and the Foundations of Mathematical Formalism

George Lotishvili

Associate Professor, East European University

Tbilisi, Georgia

`g_lotishvili@eeu.edu.ge`

2026

Zenodo DOI: 10.5281/zenodo.21325652

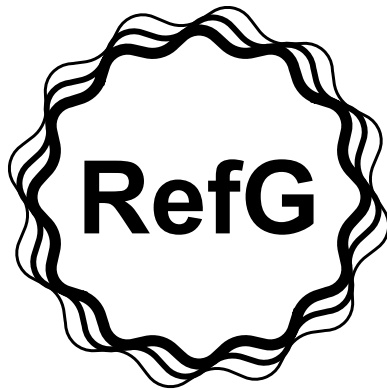
Abstract

This monograph develops the systematic ontological architecture of **Refractive Gravity** (RefG), a unified physical framework in which gravitation, discrete matter, and the operational structure of measurability emerge from one universal pre-spatial foundation. Its working Genesis scenario begins at a centerless initial boundary common to the entire foundation, assigned $t = 0$ and $\tau = 0$, where the first minimally distinguishable relations appear. Nonlinear modes subsequently localize as oscillon seeds; overlap of their resonant fields and thresholded resonant coupling can increase network connectivity, while stable matter forms only through later locking. This causal ordering is internally consistent under the tested assumptions, but the transition action, actual topology, coupling dynamics, and initial thermal state remain open. Locally confined nonlinear wave configurations—*oscillons*—induce a static pressure deficit in the surrounding medium through Bernoulli-type dynamics ($-\Delta P \propto \rho_{\text{dyn}}$), registered externally as active gravitational mass. Gravitational attraction and light deflection emerge as universal refraction across this inhomogeneous pressure landscape; on the static weak-field first post-Newtonian branch, the biconformal mechanism yields $\beta = \gamma = 1$ and recovers the classic predictions of General Relativity. Across astrophysical scales, the framework develops a conditional finite-core map for compact interiors, connects a macroscopic polarization potential to MOND and the asymptotic Baryonic Tully–Fisher relation, and reports a fixed-input comparison with 2,788 SPARC measurements, with an RMS scatter of 0.141 dex and a mean residual of -0.012 dex. Cosmologically, the selected expansion–relaxation branch distinguishes growth of the foundation link scale a from contraction of the material standard p , with observable scale $A = a/p$ and a finite process history whose duration remains to be derived. Type Ia supernova time dilation and early-galaxy maturity are separate tests of this branch. At microscopic scales, a candidate three-dimensional volumetric resonator and population-locked phase synchronization provide a spectral map for discrete particle modes and a candidate C_3 /Koide construction, while $Lk = Wr + Tw$ organizes geometric candidate readings of spin-1/2, Dirac $g = 2$, electric charge, and color confinement.

Contents

Preface	4
Introduction	4
Operational Foundations and Three Fundamental Principles	5
Lorentz Invariance and Historical Precedents	5
Seven Core Theoretical Contributions	5
Historical Lineage and Priority Boundaries	7
1 The Ontological Foundation	7
1.1 A Single Foundation and Its Manifested Medium: Substance, Void, and the Dynamics of Existence	7
1.2 From Single Existence to Multiplicity: The Emergence of Measurability and Operational Space	8
1.3 Nodal Transmission: Waves, Time, and Self-Calibration	10
1.4 Local Collective Cadence and Resonant Locking	11
1.5 Oscillons, Pressure Deficits, and the Origin of Mass	12
2 Gravity as Refraction	13
2.1 Three Simultaneous Operational Shifts	13
2.2 Quadratic Coupling and the Factor of Two	15
2.3 Refractive Attraction, Inertia, and the Absence of Friction	16
2.4 Local Unobservability and the Principle of Relativity	17
2.5 Historical Context and the Role of RefG	18
3 Three Scales and Cosmological Horizons	19
3.1 The Local Scale: Compact Objects, Internal State, and External Readout	19
3.2 The Galactic Scale: Vortical Flow and MOND	20
3.3 The Cosmological Scale: Expansion of the Manifested Domain, Foundation Relaxation, and Two Descriptions of Time	23
3.4 Quantum Randomness and Two-Level Time	26
4 Oscillons and the Origin of Matter	26
4.1 Many Patterns in One Ocean	26
4.2 Localized Confinement: Massive and Massless Modes	27
4.3 Self-Forming Resonators and the Two-Level Filter	28
4.4 The Global Resonator: Unity of Micro- and Macro-Spectra	29
4.5 Discreteness and the C_3 Generation Symmetry	30
5 Topology and Quantum Numbers	31
5.1 Phase, Orientation, and Topological Quantization	31
5.2 Spin and Fermionic Statistics	32
5.3 Charge, the Photon, and the Electromagnetic Channel	33
5.4 Color Confinement and Three Generations	33
5.5 Geometric Map of the Standard Model	34
5.6 Fundamental Constants: α and the Gravitational Hierarchy	35
6 Epilogue: Boundaries of Knowledge	36
6.1 Why One Foundation and One Manifested Medium?	36
6.2 Three Operational Boundaries	37
6.3 Methodological Principles	38
6.4 Where This Program Could Die (The Falsification Map)	38

Conclusion	39
Current Status of the Research Program	40
References	42



Preface

This monograph presents an ontological account of Refractive Gravity (RefG), unifying diverse lines of theoretical inquiry into a single, coherent physical vision. RefG is a research program under stepwise, disciplined development. The central aim of this text is to provide the reader with a deep, intuitive picture of the fundamental architecture of the Universe; the underlying computational formalism and executable verification gates are preserved in the public scientific archive on Zenodo [Lotishvili, 2026a,b].

For the current status of the hypotheses, formal proofs, and empirical boundaries discussed throughout this work, the reader is referred to the concluding section, “Current Status of the Research Program.”

Introduction

Modern fundamental physics rests upon a monumental, yet internally divided, architecture. On the one hand, Albert Einstein’s General Relativity describes gravitational interaction as the dynamic geometry of a smooth, four-dimensional pseudo-Riemannian spacetime; on the other, Quantum Field Theory represents matter and gauge forces as operator-valued excitations on a fixed, predominantly flat Minkowski background. Both pillars enjoy unprecedented levels of empirical confirmation, yet their internal ontological foundations remain fundamentally incompatible: gravity rejects a static background and dynamically generates the arena itself, whereas quantum mechanics, to preserve unitarity and its probabilistic structure, a priori requires a pre-determined, external temporal parameter.

At the interface of this profound disconnection, contemporary physics confronts a cluster of unresolved foundational crises:

1. **The Singularity Problem:** At the cores of black holes and at the cosmological origin, the continuous geometry of classical spacetime terminates in infinite curvature, signaling a breakdown of physical predictability.
2. **Astrophysical and Cosmological Anomalies:** Galactic rotation curves, the Radial Acceleration Relation (RAR), and the Baryonic Tully–Fisher Relation (BTFR) are conventionally accommodated by postulating dark matter halos, while the accelerated expansion of the Universe is attributed to dark energy (the cosmological constant Λ), whose theoretical vacuum expectation value exceeds observational limits by 120 orders of magnitude.
3. **Structural and Hierarchical Puzzles:** The origin of quantum measurement randomness, the physical nature of the arrow of time, the pronounced mass hierarchy among charged leptons and quarks, the unexplained values of dimensionless coupling constants (such as the fine-structure constant $\alpha \approx 1/137$), and the extreme weakness of gravity relative to the electroweak scale ($M_{\text{Pl}}/m_e \sim 10^{22}$) remain arbitrary, disconnected inputs in the Standard Model.

The history of physics demonstrates that such persistent, isolated paradoxes typically indicate the absence of a deeper ontological substrate. Numerous grand unifications have sought to bridge this divide: Albert Einstein spent his final decades attempting to subsume electromagnetism and matter into an extended spacetime geometry; John Archibald Wheeler explored the nexus of geometry and information through the guiding principles of *Mass without Mass* and *It from Bit* [Wheeler, 1955, 1990]; Andrei Sakharov interpreted gravity as an elastic resistance (“induced gravity”) emerging from the quantum fluctuations of the vacuum [Sakharov, 1967]; Ted Jacobson derived Einstein’s field equations directly from the thermodynamic equation of state of local causal horizons [Jacobson, 1995]; Erik Verlinde formulated gravitation as an emergent holographic entropic force [Verlinde, 2011]; and Roger Penrose formalized the inescapable connection between global causal structure, gravitational collapse, and spacetime singularities [Penrose, 1965].

While these frameworks employ distinct mathematical vocabularies, they share a singular, foundational question: **Is it possible that the entire tapestry of physical phenomena—space, time,**

gravity, and the particle spectrum—is the dynamic manifestation of a single primary ontological substrate?

This monograph answers this question in the affirmative, presenting the unified conceptual and ontological framework of **Refractive Gravity** (RefG).

Operational Foundations and Three Fundamental Principles

The standard language of physics typically begins with an established measuring apparatus: smooth coordinates, a continuous time axis, and operator fields are treated as axiomatic primitives. RefG grounds its inquiry in the strict operationalism of Percy Bridgman [Bridgman, 1927]: *Before speaking of space, time, or mass, what physical process generates the local, distinguishable differences upon which rulers and clocks function?*

In 1905, Einstein dismantled the concept of absolute time by operationally defining simultaneity via clocks and light signals [Einstein, 1905]. RefG pushes this operational critique to a more primordial level: **What physical mechanism constructs the stable clock, the material particle, and the propagating wave signal upon which all such measurements fundamentally depend?**

In response to this question, RefG formulates three fundamental postulates:

1. **Monistic Ontological Foundation:** Nature possesses strictly one universal, pre-spatial, and pre-temporal foundation. Its resonant self-distinction generates a manifested medium whose phase, pressure, shear, rotational, and topological modes are operationalized as elementary fields, forces, and particles.
2. **The Oscillon Nature of Matter:** A stable massive material unit is an *oscillon*—a locally confined, self-sustaining nonlinear wave configuration of the manifested medium. Massless carrier modes are the freely propagating radiation waves of the same medium. The localized oscillatory action of an oscillon generates a persistent, long-range static pressure-deficit profile in the surrounding substrate.
3. **Gravity as Effective Refraction:** Oscillon matter establishes an inhomogeneous pressure-deficit landscape throughout the medium. This deficit alters the effective geometry, causing the wavepaths of light and matter to bend refractively. Gravitational attraction is thus revealed as the physical mechanism of spacetime curvature.

Lorentz Invariance and Historical Precedents

The Michelson–Morley experiment of 1887 detected no “aether wind” from the Earth’s orbital motion, placing stringent constraints on 19th-century models of a rigid, matter-independent “luminiferous aether” [Michelson and Morley, 1887]. The RefG foundation-medium belongs to an entirely different ontological category: here, measuring rods, clocks, the observer, and propagating light are themselves collective wave excitations woven from this very medium.

When an observer and their apparatus are co-emergent wave modes of the medium, uniform translational motion corresponds to a uniform phase transformation of the wavepacket. Along the Lorentz-covariant branch, this motion leaves no dissipative wake or detectable “wind.” This realization revitalizes Lord Kelvin’s (William Thomson’s) visionary intuition regarding “vortex atoms” [Thomson, 1867]: **Matter is not a foreign body inserted into a passive background—matter is the organized, self-sustaining motion of the medium itself.**

Seven Core Theoretical Contributions

This text establishes the ontological architecture of RefG, endowing the computational formalism archived on Zenodo with unambiguous physical meaning. While medium-based unifications have notable precedents—such as Winterberg’s Planck aether model [Winterberg, 2002]—RefG places particular emphasis on the operational genesis of measurability and population-level resonant locking.

The core theoretical findings of this work are distilled into seven principal points:

1. **The Birth of Time and a Common Oscillon Cadence:** In the working origin scenario of RefG, time begins at a centerless initial transition common to the entire foundation, which produces the first minimally distinguishable relations. Physical duration and event order are undefined on the pre-origin side of this boundary. Nonlinear modes localize as oscillon seeds only within the relational layer that has already formed, while stable material oscillons arise through later locking. Their local collective cadence, $\Omega(x)$, is established by nonlinear mutual locking and resonant synchronization ($\nu_i = n_i \Omega(x)$). Gravitational time dilation is the coordinated reconfiguration of this local resonant network in response to a change in foundation pressure.
2. **Two-Channel Mass Accounting:** The theory rigorously decouples two physical quantities: the oscillon's internal topological state (its invariant internal energy store) and its externally read gravitational mass, which represents the volume-integrated pressure deficit induced in the surrounding medium.
3. **Biconformal p/p^2 Scaling and the Factor-of-Two Deflection:** Along the first-order biconformal branch, operational length, externally read mass, and clock rates scale linearly with substrate pressure (p), whereas the coordinate propagation of light scales quadratically (p^2), rooted in the differing geometrical nature of network nodes versus interconnecting links. In the weak-field regime, this mechanism exactly reproduces the metric predictions of General Relativity, yielding the double deflection of light and Shapiro time delay (PPN $\beta = \gamma = 1$). At macroscopic scales, this scalar-pressure channel, augmented by the non-integrable strain of topological defects and the effective field theory (EFT) framework of Fierz–Pauli and Deser [Deser, 1970], closes into the full Einstein–Hilbert action.
4. **The Koide C_3 Phase Geometry:** In the three-phase mass map of charged leptons (e, μ, τ), the mass angle $\theta = h/9$ represents the reduced coordinate of a ninth-order holonomic return cycle. On the selected candidate branch, the first non-trivial topological return gives $h = 2$ and $\theta = 2/9$, while internal vibrational-channel equipartition is associated with Koide's dimensionless modulation amplitude $A_K = \sqrt{2}$.
5. **The Nodal Origin of the Fine-Structure Constant (α):** The fine-structure constant $\alpha \approx 1/137$ is interpreted as the dimensionless impedance ratio of the substrate's electric-flux and transverse-photon response channels, remaining invariant under local pressure transformations.
6. **A Candidate Genesis, Physical Expansion, and the Global Slowing of Time:** The selected scenario begins with a centerless initial boundary common to the entire foundation, assigned $t = 0$ and $\tau = 0$. The first distinguishable relations and the first physical ordering of events arise there. Resonant fields extending from oscillon seeds within the already formed relational layer overlap; the resonant coupling produced when a physical distinguishability threshold is crossed increases the connectivity and operational extent of the network. Stable matter, radiation, and long-term relaxation belong to later phases. The internal W3-37–W3-39 reports test only three narrow conditions: the selected event ordering contains no closed causal cycle; connectivity without a unique center is possible on a finite periodic graph; and a conditional post-origin ledger of energy-transfer signs closes exactly. They do not derive the physical trigger of the first transition, the actual topology of the Universe, the propagation and coupling dynamics of resonant fields, or the initial thermal state. A separate post-origin cosmological branch considers simultaneous growth of the foundation link-length scale and contraction of material standards, combined in the operational scale $A = a/p$. The history measured by present-day clocks spans approximately 13.8 Gyr; process history is also finite on the selected branch, while its exact duration remains to be derived.
7. **Two Readouts of a Single Spectral Operator—A Central Hypothesis:** The short-wavelength eigenfunctions of a unified volumetric Helmholtz/Laplace-type spatial resonator are

read as microscopic particle states, whereas its long-wavelength modes are read as candidate organizers of the cosmic web and galaxy-cluster structure.

Historical Lineage and Priority Boundaries

Every component of RefG is anchored in the established heritage of theoretical physics:

- The p/p^2 biconformal scaling shares conceptual kinship with the Polarizable Vacuum (PV) frameworks of Dicke and Puthoff [Dicke, 1957, Puthoff, 2002]; in RefG, it arises from the discrete ontology of a nodal network and two-channel mass accounting rather than phenomenological dielectric permittivity.
- The three-phase geometric reading of Koide’s formula extends the pioneering work of Brannen and Foot [Brannen, 2010, Foot, 1994], while the holonomic origin of $2/9$ resonates with Shulga’s compact family cycles [Shulga, 2026]. RefG unifies this structure with θ -blindness arguments and the C_3 tetrahedral symmetry of a 3D volumetric resonator.
- Cosmological vacuum-pressure relaxation builds upon the q -theory of Klinkhamer and Volovik and its tendency toward saturation [Klinkhamer and Volovik, 2008], while the motivation for three spatial dimensions is related to studies of topological knotting and vortex-line stability [Whitrow, 1955, Berera et al., 2017].

The defining contribution of this monograph is the synthesis of these disparate threads into a single, logically connected research program: tracing the path from the operational birth of measurability and oscillon mass to galactic vortex dynamics and cosmological horizons.

The text invites the reader to follow this chain and see how a unified physical intuition unfolds into a broad panorama of contemporary fundamental physics.

1 The Ontological Foundation

1.1 A Single Foundation and Its Manifested Medium: Substance, Void, and the Dynamics of Existence

Modern fundamental physics describes the Universe as an assembly of mutually independent fields. In the Standard Model, each field is equipped with its own internal gauge symmetry, intrinsic degrees of freedom, and coupling parameters. Refractive Gravity (RefG) replaces this pluralistic ontology with a single, uncompromising monistic principle: **Nature possesses strictly one universal, pre-spatial ontological foundation.** Its resonant self-distinction generates a measurable, multi-channel medium. What we observe as distinct fundamental fields, gauge interactions, and elementary particles are the topological, phase, pressure, shear, and rotational modes of this single manifested medium. Throughout this monograph, the term “foundation-medium” denotes this self-distinguished, physically manifested state of the primary substrate.

This monistic vision possesses deep historical roots in theoretical physics. As early as 1870, William Kingdon Clifford formulated the prophetic hypothesis that matter consists of small, propagating ripples of spatial curvature [Clifford, 1876]. RefG endows this geometric intuition with a concrete physical mechanism: the self-distinguished medium supports not only geometric curvature responses, but also pressure, shear, hydrodynamic, and resonant channels.

A prominent theoretical precedent for the emergence of diverse particle and gauge sectors from a single substrate is the *string-net condensation* framework of Michael Levin and Xiao-Gang Wen [Levin and Wen, 2005]. In their model, collective many-body entanglement within a simple bosonic system naturally gives rise to both gauge bosons (photons, gluons) and fractional fermionic excitations (electrons, quarks). While the Levin–Wen model presupposes a pre-existing spatial lattice, RefG grounds this unifying capacity in the dynamic self-distinction of a single, pre-spatial foundation.

Conceptualizing the Universe as a unified continuum requires two asymptotic dynamical bounds:

1. **The Absolute Maximum of Substance:** A saturation limit characterized by maximal pressure, mass-energy potential, and operational nodal density.
2. **The Zero (Vacuum) Bound:** A limit in which interaction and physical measurability vanish entirely.

This “void” is not an abstract metaphysical nothingness, but the operational limit where physical distinguishability ceases. Objective physical reality unfolds as the bounded, nonlinear oscillatory dynamics between these two poles, wherein stable local units of the foundation—“nodes of existence”—are formed.

The working background $\varphi = 0$ is the quiescent yet physically active state situated between these two extrema. At the limit of zero distinguishability, nodal interactions and measurable traces completely vanish; $\varphi = 0$ designates the unperturbed baseline state of the manifested medium.

The ontological unity of the manifested medium is accompanied by a rich multi-channel capacity. The medium can compress (volumetric pressure), twist (topological torsion), establish vortical circulation, and support transverse wave excitations. A hydrodynamic analogy provides useful intuition: a single fluid concurrently supports isotropic hydrostatic pressure, vortex circulation, shear stress, and surface waves. Similarly, the RefG manifested medium integrates phase, pressure, shear, and topological responses, providing the dynamic substrate upon which electric charge, spin, and the particle spectrum are constructed.

The unperturbed state ($\varphi = 0$) represents the baseline vacuum regime. The foundation is omnipresent; what varies locally is its degree of excitation, the operational density of distinguishable nodes, and the coupling strength between them. A drop in pressure, or rarefaction, corresponds to an increase in the unmeasured operational intervals separating nodes.

Here, “pressure” serves as the effective state variable of energy density in RefG. The baseline $\varphi = 0$ allows deviations in both directions: $\varphi > 0$ corresponds to a compressed, high-pressure branch, while $\varphi < 0$ denotes a rarefied, deficit branch. The entire gravitational readout in RefG is constructed upon the gradient profile of this deficit branch ($\varphi < 0$). Classical Bernoulli pressure serves as the direct physical analogue: for an external observer, a local shift in φ is simultaneously registered as gravitational time dilation, coordinate retardation of light, operational length contraction, and effective gravitational mass.

1.2 From Single Existence to Multiplicity: The Emergence of Measurability and Operational Space

Under the working postulate of RefG, physical change, event order, duration, and geometry are not yet assigned to the foundation’s initially undifferentiated state. In this picture there is no elapsed time before the birth of the Universe: birth establishes the first physical event and the origin of temporal reckoning.

The most coherent working origin scenario adopted here is a centerless initial transition common to the entire foundation. It is one initial condition with no distinguished spatial center. It does not describe a signal propagating through an already existing space, because physical distance, propagation speed, and an external vantage point are undefined on its pre-origin side.

At this boundary the foundation first produces minimally distinguishable relations. The physical ordering of events and both temporal measures begin there, with $t = 0$ and $\tau = 0$. This first layer of distinguishable connections is the minimal network of existence; material particles and stable oscillons have not yet formed.

Once these relations exist, some nonlinear modes localize as oscillon seeds. Resonant fields generated by those seeds propagate over the existing relational layer and overlap. In the working scenario, when a physical distinguishability threshold is crossed, the resulting resonant coupling joins operationally separated regions into one connected network. The first distinguishable relation therefore precedes this coupling. Local locking and network-wide connection may be partly contemporaneous; their actual ordering must follow from the dynamics of the resonant fields and the physical coupling threshold.

The most natural candidate form of the Universe is an interconnected network with neither a boundary nor a distinguished center. An exact finite periodic-graph construction shows only that finiteness, connectivity, and the absence of a unique center are mathematically compatible. It is an existence witness, not a derivation of the Universe’s dimension or topology.

Energy in the initial coherent state may be redistributed among oscillon seeds, their resonant fields, freely propagating waves, a thermal sector, and the collective deformation of the foundation. In the conditional post-origin local ledger of W3-39, assuming a continuity law and a moving boundary, the energy change separates into a volume source, outward boundary flux, a possible central term, and the energy swept into the enlarging domain. Antisymmetric transfers among predeclared sectors cancel in the total. The actual genesis energy law remains open. At this stage *melting* is only a working metaphor: whether conversion releases or absorbs energy depends on phase energies and the equation of state.

In this scenario the first collective change initiates the foundation’s self-distinction. Operational space becomes measurable when a stable relation between nodes reaches the minimum distinguishability scale ℓ_* . The interval between any two distinguishable nodes is the physical embryo of space; signal delay across their relation produces measurable duration, while the local energetic strain of differentiation is the ontological embryo of mass. Externally read gravitational mass is defined later by integrating the deficit profile that this strain induces in the surrounding foundation.

Physical meaning invariably requires comparison. At the ontological origin, three indivisible elements emerge:

- **Node:** A locally distinguishable, discrete state of the foundation;
- **Trace:** The energetic and phase alteration imparted by a node to the surrounding medium;
- **Relation:** The operational capacity to compare one local state with another.

From this foundational triad unfolds the entire operational network that macroscopically manifests as space, time, and matter.

A completely undistinguished foundation possesses a single, indivisible identity. Its resonant self-distinction generates the first persistent difference, marking the birth of quantity and counting. Distinguishable manifestations are registered as separate physical units without the foundation fragmenting into disconnected parts—multiplicity consists of the stable, mutually distinct states of one fundamental identity.

The primary unit of measurement is not an isolated object, but a preserved relation. In an undistinguished state, no second entity exists for comparison. Self-distinction yields two distinct states of the same foundation and establishes the primary link between them; here begin counting, spatial extension, and the relative ordering of events.

In information-theoretic terms, the first preserved distinction is analogous to a bit: a distinguishable trace is registered as “1”, and its absence within a given relational frame as “0”. Complex arrangements of these binary states generate rich macroscopic structures, just as the mutually distinct states of the foundation generate the diversity of the observable cosmos.

The concept of a single object existing at different locations at different times illustrates this logic. If the temporal labels along its worldline are removed, a single identity leaves traces across multiple locations. RefG elevates this intuition to the pre-spatial domain: persistent self-distinctions of the foundation construct the very locations that can subsequently be counted and compared. The atemporal status of the foundation and the metric age of the observable Universe thus belong to distinct conceptual levels.

This perspective is crucial for understanding the nature of the observer. Observers and their measurement instruments are themselves complex, composite wavepackets built from this nodal network. This deepens Percy Bridgman’s operationalism, which asserts that physical concepts are defined solely by the operations required to measure them [Bridgman, 1927]. RefG provides this operational philosophy

with an ontological substrate, investigating the physical conditions that make measurement operations possible in the first place.

The observable Universe is not matter placed inside an empty geometric container; it is the ongoing process of self-distinction within the foundation-medium. Operational space emerges along the traces of this distinguishability. Beyond these traces, the foundation may persist, but as a physical object it manifests only where it leaves an operational, measurable imprint.

The absolute separation between nodes cannot be measured directly, because no observer possesses a measuring rod external to the medium. What is measurable is strictly the operational trace of the inter-node interval. Operational space is born when the relation between at least two nodes becomes physically distinguishable. Below this threshold, variation remains an unmanifested internal state; above it, it registers as shifts in clock rates, operational sizes, mass readouts, and signal propagation.

RefG posits a dynamic lower bound of minimal operational distinguishability, denoted by ℓ_* :

- At scales smaller than ℓ_* , no independent measurable relation can form (information is indistinguishable);
- At the scale ℓ_* , the first stable, physically distinguishable record emerges;
- At scales larger than ℓ_* , inter-node intervals are operationalized as changes in pressure, clock rates, and signal propagation.

The physical rationale for this bound is that an infinitely fine record would lack dynamical stability. While the Planck length $\ell_{\text{Pl}} = \sqrt{\hbar G/c^3}$ is a dimensional scale constructed from fundamental constants, ℓ_* represents the dynamical threshold for stable record formation. Their potential coincidence depends on the microdynamics of the foundation.

Consequently, the ontological origin of the Universe is not a pre-existing manifold of geometric points. Prior to the emergence of inter-node intervals, there are neither distinct points nor spatial extension. Such a state cannot even be termed a “single point,” as a point is a concept belonging to an already realized geometry; it is a completely undifferentiated primary foundation-state. Multiplicity and geometry emerge only when an asymmetry generates a stable trace, rendering inter-node relations operationally readable.

Here lies the physical essence of gravity: when local coupling between nodes weakens (pressure falls), this state is simultaneously registered by an external observer as clock retardation, contraction of operational scale, and signal delay. This coordinated triad of physical responses is precisely what macroscopic observers perceive as a “gravitational field.”

1.3 Nodal Transmission: Waves, Time, and Self-Calibration

The foundation’s first collective change creates a minimal relational layer. Each node is a functionally distinguishable state that preserves the trace of a physical change and can relay it to adjacent nodes. Subsequent transmission, together with resonant coupling produced by overlapping resonant fields of oscillon seeds once the distinguishability threshold is crossed, joins this layer into an operational network. Spatial proximity and operational distance emerge from the stable organization of these physical couplings.

A wave in this network is the sequential transmission of state and phase. Light is a freely propagating transverse phase trace, whereas an oscillon is a locally confined, self-sustaining configuration of the same transmission mechanism. When matter moves, this confined wavepacket is transferred from node to node, reconstructing its phase integrity at each successive location. Light and matter are thus the propagating and localized modes of a single medium.

In the working origin scenario, the first foundation-wide self-distinction creates the first ordering of events and begins time. Every subsequent transmission within the formed network extends that order. Recurrent phase relations establish a process cadence, while comparisons among stable oscillon cycles make clock measurement possible. The local rate of time is the network’s collective rhythm: the rate at which the medium relays state and an oscillon completes its internal cycle.

Thermal insulation in double-pane windows provides a helpful analogy. Rarefying the intermediate gas reduces molecular collisions and slows the rate of heat conduction. RefG extends this intuition to the nodal network, linking the transmission rate directly to nodal density and coupling strength.

The observer's body, clocks, rulers, light, and matter are all modes of this same medium. In a deficit region, nodal density, transmission rates, oscillon cycles, and operational sizes undergo a coordinated shift. Local measuring devices reconfigure synchronously with their environment; consequently, an observer using local standards measures normal clock rates, standard dimensions, and the invariant local speed of light (c). Comparing processes that have traversed different environments, however, reveals objective differences in accumulated time, operational size, and mass.

Along the selected Lorentz-covariant low-energy branch, this transmission is isotropic, non-dissipative, and strictly respects Lorentz symmetry: the local speed of light remains constant, and uniform motion leaves no frictional wake.

1.4 Local Collective Cadence and Resonant Locking

The birth of time and the formation of a local collective cadence are distinct stages. In the working scenario, time begins with the first foundation-wide self-distinction. A local common cadence forms later, when the intrinsic frequencies of stable material oscillons settle into fixed, persistent ratios. In that collective state, the temporal rate is governed by the internal coordination and nonlinear mutual locking of the local network, in a manner analogous to Kuramoto-type synchronization.

At a deeper ontological level, the foundation possesses neither a preassigned tone nor a ready-made clock. After the connection network has formed, the manifested medium may behave like an acoustic instrument—a vibrating volumetric resonator whose connection operator and boundary conditions determine the admissible modes. The three-dimensionality of this resonator belongs to the candidate physical map of RefG.

The physics of a bell offers a vivid illustration: the acoustic signature of a bell is not determined by isolated, arbitrary frequencies, but by the normal modes of a continuous elastic body resonating in collective harmony. Similarly, the frequencies of elementary particles are coupled through mutual balance within the energetic reservoir of the foundation. The universal foundation sets the allowed modal arena, while particle modes mutually equilibrate across this arena to form a stable, resonantly locked system.

Denoting the local collective cadence by $\Omega(x)$, the frequency spectrum of stable oscillons is expressed as:

$$\nu_i(x) = n_i \Omega(x), \quad (1)$$

where n_i is a fixed, dimensionless structural factor characteristic of a specific particle mode. In general, n_i is non-integer, so particle mass and frequency spectra do not reduce to simple one-dimensional overtones. Within the candidate three-dimensional volumetric-resonator map, they correspond to a complex, slightly anharmonic eigenspectrum, examined below through the C_3 symmetry of charged leptons.

The relation $\nu_i(x) = n_i \Omega(x)$ preserves individual mode identities while locking every mode to the shared local cadence $\Omega(x)$. Consequently, the frequency ratios:

$$\frac{\nu_i(x)}{\nu_j(x)} = \frac{n_i}{n_j} = \text{const} \quad (2)$$

remain strictly invariant under resonant locking. Any oscillatory rhythm incompatible with the collective network loses phase coherence and cannot persist as a stable particle mode.

In a region of low pressure (gravitational deficit), the collective cadence $\Omega(x)$ of the entire local network decreases. In a dense, high-pressure substrate, inter-node coupling is robust and $\Omega(x)$ has a higher value; in a rarefied substrate, coupling weakens and $\Omega(x)$ slows down:

- The retardation of inter-node signal transmission,

- The reduction in substrate pressure-energy density, and
- The reduction of the collective cadence $\Omega(x)$

represent three distinct operational readouts of a single ontological reality; within the formal mathematical chain, each enters exactly once.

When $\Omega(x)$ shifts locally, all local clocks, atomic transitions, and particle processes scale synchronously. Consequently, a local observer cannot detect any variation in their own rate of temporal passage. Under this collective rescaling, all dimensionless ratios—including the fine-structure constant α and mass ratios m_i/m_j —remain strictly invariant, reflecting the internal geometric-topological proportions of the network rather than its absolute rate.

The historical precursor to this relational approach is Ernst Mach’s principle [Mach, 1883]. RefG gives that principle a local, dynamical form: the common cadence of time is governed directly by the collective state of the local resonant network, while the influence of distant masses must be encoded in that local state. Relationality and local causality are therefore properties of the same local network.

In the RefG framework, the fine-structure constant $\alpha \approx 1/137$ is interpreted as the dimensionless impedance ratio of the substrate’s electric-flux and transverse-photon response channels, remaining invariant under local shifts in substrate pressure.

Now consider a wavepacket propagating between two pressure regimes of the foundation, for example from a gravitational well to a distant observer. Its observed frequency is computed from the effective metric. In the ontological language of RefG, the local cadences of source and receiver are compared through phase translation. On a cosmological background the operational scale is $A = a/p$, so spectroscopic and metric redshift follow the same ratio,

$$1 + z = \frac{A_0}{A_{\text{em}}} = \frac{a_0/p_0}{a_{\text{em}}/p_{\text{em}}}. \quad (3)$$

The signal’s phase relation is translated into the local $\Omega(x)$ cadence in each region. A compatible signal locks into the new resonant network, whereas an incompatible signal dephases or scatters and redistributes its energy among the pressure–energy channels of the foundation.

In summary, the rate of time in RefG is the ontological state of the local resonant network. When substrate pressure shifts, the network undergoes a collective rescaling: all compatible processes scale synchronously, leaving dimensionless ratios strictly invariant. An independent, abstract universal time possesses neither operational utility nor physical meaning: every physical measurement is executed strictly from within the network itself.

This principle establishes a rigorous empirical falsification criterion: **If any dimensionless ratio changes under a local pressure shift—for example, if a reliably established nonzero $\Delta\alpha/\alpha$ is found—the postulate of a population-locked collective cadence is falsified.**

1.5 Oscillons, Pressure Deficits, and the Origin of Mass

What traditional physics designates as an “elementary particle”—whether an electron, proton, or quark—is recognized in RefG as an **oscillon**. An oscillon is a locally confined, self-sustaining, nonlinear topological wavepacket (a soliton-like state) supported by the foundation-medium. It is not an isolated, solid corpuscle distinct from the background; rather, it is a highly organized, coherent oscillatory state of the medium itself.

This concept continues the profound lineage of Lord Kelvin’s (William Thomson’s) “vortex atoms” [Thomson, 1867]. While 19th-century mechanical aether models were superseded, Kelvin’s core insight—that particles are dynamic, self-confined topological modes of a continuous medium—remains a foundational concept that oscillon physics formalizes within modern nonlinear field theory and topology.

When an oscillon oscillates intensely, rotates, or confines energy within its localized geometry, the static pressure of the surrounding foundation-medium decreases. This behavior is illuminated by Daniel Bernoulli’s hydrodynamic principle [Bernoulli, 1738]: **An increase in internal dynamic intensity**

(velocity, oscillation amplitude) induces a corresponding drop in local static pressure. On the weak, static gravitational branch of RefG, this hydrodynamic analogue corresponds to the schematic pressure-energy relation:

$$\Delta P \propto -\rho_{\text{dyn}}. \quad (4)$$

Through its continuous oscillatory action, an oscillon imprints a stable static pressure-deficit gradient upon the surrounding foundation. Within this framework, this pressure deficit is nothing other than the **direct external profile of gravitational mass**: integrating this deficit over all space yields the total gravitational mass of the system. The more intense the oscillon’s internal dynamics, the deeper the induced pressure deficit and the stronger the resulting refractive gradient. Simultaneously, the oscillon’s internal topological energy store is accounted for as a separate, decoupled physical channel.

The core of an oscillon is localized, but its oscillatory trace extends throughout the connected medium. The component phase-locked to the source establishes a static gravitational profile, requiring zero net outward energy flux. When the source undergoes rapid dynamic reconfiguration, the coherent oscillatory component released from this channel propagates outward as a gravitational wave, carrying energy and momentum.

The local depth of the deficit and the asymptotically measured total mass of a system are distinct physical quantities. The former characterizes central gradient and compactness; the latter integrates the active core, substrate dressing, binding energy, and radiated flux into a single asymptotic balance. Consequently, following a binary merger, the central deficit deepens while the total externally read mass decreases by the exact amount of radiated gravitational-wave energy.

This perspective shares structural parallels with John Archibald Wheeler’s concept of *Mass without Mass*, in which a “geon” represents a massive entity bound entirely by the self-energy of its gravitational and electromagnetic fields [Wheeler, 1955]. While classical Wheeler geons are unstable to radiative decay, RefG stabilizes oscillons through topological charge conservation, resonant phase locking, and the topological knot invariant $Lk = Wr + Tw$.

Consequently, the term “effective refractive mass” (m_{eff}) designates the externally read gravitational mass of an individual oscillon within a specified background medium. Under adiabatic background variations, the oscillon’s topological structure, internal energy, and phase ratios preserve their identity, while its operational scale and externally read mass adapt to the new substrate state. Rapid reconfiguration represents an independent dynamical process that redistributes energy into radiation channels. The total asymptotic mass of an extended system requires a volume integration and is not simply the linear sum of uncoupled single-oscillon m_{eff} values. Through this decoupling, the origin of mass is revealed as the establishment of the medium’s refractive response.

At this foundational level, **gravitational attraction is the refractive bending of one oscillon’s wavepacket within the pressure deficit generated by another**. A relaxed string provides an intuitive picture of a gravitational well: locally diminished pressure-energy weakens the effective wave propagation stiffness of the substrate. The deficit profile defines the effective geometry, and any matter wavepacket bends along geodesics of minimal optical path length according to the eikonal equation.

This defines the central dynamical reciprocity proposed by RefG: **An oscillon creates a pressure deficit through its internal oscillation, while this deficit determines its effective mass, operational scale, and trajectory through the medium**. Matter dictates the pressure state of the medium, and the altered medium dictates the refractive geometry of matter.

2 Gravity as Refraction

2.1 Three Simultaneous Operational Shifts

As established in the preceding chapter, the inter-node separation within the foundation-medium is not directly accessible to external measurement; an observer can access only its operational, physical traces. Around an oscillon, this trace manifests as a localized depression in the pressure-energy state of the substrate: distinguishable nodes are rarefied, and the informational and wave coupling between them

is attenuated. For an external, macroscopic observer, this single ontological state manifests as three simultaneous, coordinated operational shifts:

1. **Deceleration of the Coordinate Speed of Light:** In a pressure-deficit zone, nodes are rarefied and the rate of inter-node transmission drops. Consequently, the propagation of light and all other massless wave signals is retarded in coordinate description. The locally measured speed of light remains strictly invariant (c) for a local observer; what changes is the effective propagation rate evaluated in the coordinate language of a distant observer.
2. **Contraction of Oscillon Operational Size:** While the oscillon’s intrinsic topological form (its nodal winding configuration) remains invariant, it is populated by a reduced volumetric density of interacting nodes within the rarefied substrate. In external observational description, this is registered directly as a contracted operational size, manifesting as a modified effective mass readout.
3. **Retardation of the Local Resonant Clock Cadence:** Real physical clocks are composed of oscillon systems (atoms, molecules) whose internal tick rate depends directly on local nodal coupling. In a rarefied environment, this coupling weakens, decelerating the clock rate in coordinate time relative to an asymptotic observer. Locally, the clock continues to measure its own cadence normally and invariantly.

A physical analogy is provided by double-pane vacuum insulation: evacuating the intermediate space reduces molecular collision frequency and suppresses the rate of thermal transport. In the RefG framework, this corresponds to the relation between nodal coupling strength and coordinate clock speed.

These three shifts are distinct operational readouts of a single underlying pressure deficit. Crucially, they scale with substrate pressure according to different powers, establishing the foundation of biconformal geometry.

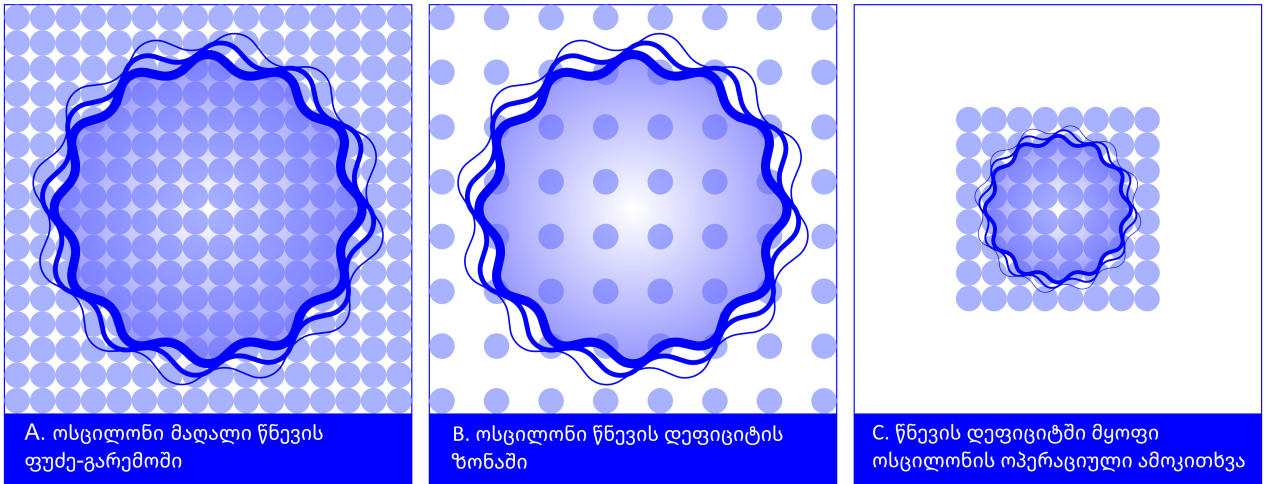


Figure 1: **Operational readout of an oscillon within a pressure-deficit zone.** (A) In an unperturbed, high-pressure foundation-medium, an oscillon resides in a dense nodal network; inter-node coupling is robust and local processes proceed at maximal cadence. (B) Within a pressure-deficit zone, substrate nodes are rarefied; operational inter-node distance increases, transmission rates decrease, and local temporal passage decelerates. (C) In external readout, an oscillon in a pressure deficit is registered with a slowed clock rate, contracted operational size, and shifted effective mass, while the internal separation does not enter as an independent direct observable.

Nodal rarefaction does not generate a “secondary space.” The inter-node interval is not an independent physical substance; macroscopic “void” is operationalized entirely through the language of pressure

variables. The sole objective manifestation of substrate rarefaction is the localized reduction of its pressure-energy state.

2.2 Quadratic Coupling and the Factor of Two

These three operational shifts do not scale with substrate pressure under a uniform power law. In the ontological architecture of RefG:

- **Operational size and single-oscillon effective mass** belong to the volumetric nodal capacity channel (extensive quantities);
- **The coordinate speed of light** belongs to the inter-node coupling strength and transmission delay channel (intensive quantity).

Denoting the dimensionless pressure factor by p , this biconformal scaling is formalized as:

$$\frac{L_{\text{oper}}}{L_0} = \frac{m_{\text{eff}}}{m_0} = \frac{\Omega}{\Omega_0} = p, \quad \frac{c_{\text{coord}}}{c_0} = p^2, \quad p = e^{\varphi/2}. \quad (5)$$

A similar power-law scaling appears historically in the Polarizable Vacuum (PV) models of Robert Dicke and Harold Puthoff [Dicke, 1957, Puthoff, 2002]. In RefG, these powers are assigned to distinct nodal-capacity and link-transmission channels rather than introduced through a phenomenological dielectric ansatz.

Formally, the external macroscopic readout of a single oscillon couples to the substrate pressure deficit through an exponential half-factor:

$$m_{\text{eff}} = m_0 e^{\varphi/2}. \quad (6)$$

On the deficit branch, $\varphi < 0$, confining the pressure factor to $0 < p < 1$. This relation describes the local external readout of an individual oscillon embedded within an inhomogeneous background: as φ deepens (pressure drops), its operational size and m_{eff} contract. Simultaneously, the total gravitational charge of the extended source generating the deficit is determined via a separate volumetric integration. This scaling follows directly from the theory’s first-order biconformal branch.

Here, four physical quantities must be rigorously distinguished:

1. **Oscillon internal state and proper energy** (M_{proper});
2. **Single-oscillon effective refractive mass in a given background** (m_{eff});
3. **Active source profile of the induced pressure deficit** (M_{active});
4. **Total asymptotic gravitational charge of the system** (ADM/Komar mass: Arnowitt et al. 1962, Komar 1959).

These four quantities organize into two nested accounting channels: the first distinguishes proper internal content from background-read mass; the second relates the externally read mass to the active core and substrate dressing. At the asymptotic boundary, these contributions combine into a single conserved charge, ensuring that every deficit trace is counted exactly once.

This self-consistent dynamic is captured by an intuitive physical analogy: **Imagine a crystal of colored ice dissolving in pure water.** The crystal is a locally frozen, ordered phase of the same water—just as an oscillon is a localized resonant locking of the universal foundation-medium. As the crystal melts, its dye disperses throughout the water, coloring the medium while the crystal itself shrinks in size. Energy is conserved: like the dissolved dye, localized binding energy transitions into an extended substrate pressure-energy state.

The more intensely the foundation is “colored” by this trace (i.e., the deeper the pressure deficit), the further local pressure drops, altering the externally read mass, operational size, clock cadence, and

coordinate speed of light. This is a self-regulating relaxation process: the more trace an oscillon imparts to the medium, the lower its capacity for further emission. This negative feedback provides a natural mechanism for deficit saturation, with its precise boundary governed by full nonlinear dynamics.

This p/p^2 scaling is formalized through the biconformal metric:

$$ds^2 = e^\varphi c^2 dt^2 - e^{-\varphi} d\boldsymbol{\sigma}^2, \quad \varphi = -\frac{2U}{c^2}, \quad (7)$$

where $U > 0$ is the gravitational potential of the selected exponential branch, coinciding at leading order with the Newtonian potential. In the weak-field limit ($U/c^2 \ll 1$), the metric components expand in a Taylor series:

$$g_{00} = e^\varphi \approx 1 - \frac{2U}{c^2} + \frac{2U^2}{c^4} + \cdots, \quad g_{ij} = -e^{-\varphi} \delta_{ij} \approx -\left(1 + \frac{2U}{c^2} + \cdots\right) \delta_{ij}. \quad (8)$$

On the selected static, spherically symmetric first post-Newtonian (1PN) branch, expanding this metric yields the Parameterized Post-Newtonian (PPN) coefficients:

$$\beta = 1, \quad \gamma = 1. \quad (9)$$

A complete clock cycle is operationalized by the ratio of operational size to coordinate velocity ($\Delta t \sim L_{\text{oper}}/c_{\text{coord}}$). If a pressure drop halves the operational size ($p = 1/2$) while reducing coordinate light speed by a factor of four ($p^2 = 1/4$), the period doubles—the clock runs precisely twice as slow.

Along this biconformal branch, the 1PN effects controlled by $\beta = \gamma = 1$ identically recover the classical predictions of General Relativity: the gravitational deflection of light, Shapiro time delay, gravitational redshift, and planetary perihelion precession [Bertotti et al., 2003, Will, 2014]. In particular, the Cassini radio-link experiment measured $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$.

Historically, the additional **factor of two** arising from the spatial metric component g_{ij} was the decisive contribution that elevated Einstein’s 1911 half-deflection (derived solely from temporal dilation) to the full geometric result of 1915–1916 [Einstein, 1916, Shapiro, 1964]. In RefG, this factor of two emerges naturally from the differing ontological scaling of nodal volumetric density (p) and link transmission delay (p^2).

2.3 Refractive Attraction, Inertia, and the Absence of Friction

An oscillon, as an extended wave configuration, exists within the inhomogeneous gradient of substrate pressure and coordinate light speed. On the side where the medium is denser, wave phase propagates rapidly; on the rarefied (deficit) side, propagation is comparatively retarded. By virtue of the Huygens–Fresnel principle and the eikonal equation, the wavefront bends toward the region of lower pressure and reduced coordinate velocity—undergoing **refraction**. In the static weak-field limit, the pressure deficit generated by a massive body constructs an effective optical-geometric medium, and the refractive trajectory of a matter wavepacket within this medium is perceived macroscopically as gravitational attraction.

In RefG, the classic phenomenon of Newton’s falling apple is understood as follows: the oscillon wavepackets of the atoms comprising the apple bend refractively toward the region of lower pressure and slower coordinate propagation established by the Earth’s mass. Familiar gravitational attraction is not an occult action-at-a-distance force—it is the direct, local refractive response of matter waves to an inhomogeneous medium.

Formally, the gravitational source consists of the projected pressure deficit and non-integrable strain in the foundation-medium. Oscillons, acting as topological defects of the substrate, introduce residual plastic strain into the continuum. In the geometric theory of elasticity, defect density is measured by the strain incompatibility tensor, which in four dimensions matches the algebraic structure of the Einstein tensor $G_{\mu\nu}$, while freely propagating transverse shear waves preserve volume and naturally manifest as transverse-traceless (TT) polarizations. A gravitational well is a 3D zone of depressed

substrate pressure and strain, wherein the total mass of a bound gravitational system is strictly less than the sum of its isolated constituents by the mass equivalent of the binding energy ($\Delta M = B/c^2$).

The gravitational-wave event GW150914 provides a sharp observational benchmark for this energy balance: approximately three solar masses ($3 M_\odot$) separating the initial binary mass from the final remnant were carried away as gravitational-wave energy [Abbott et al., 2016]. In RefG, two oscillatory sources merge into a single, deeper deficit profile, while the energy of their nonlinear reconfiguration radiates outward as transverse-traceless (TT) tensor waves. The quantitative benchmark of this framework is the simultaneous derivation of remnant mass, radiated energy, and complete waveform from a single initial state.

On the compact branch required for regular interior cores, the projected deficit source is registered as an effective violation of the Null Energy Condition (NEC); in RefG, this belongs to the elastic confinement channel of the foundation-medium.

RefG associates **inertia** with the dynamic deformation of an oscillon’s deficit profile during acceleration. When an oscillon is accelerated by an external force, its pressure profile develops a fore-aft asymmetry along the direction of motion; this gradient asymmetry generates a reactive resistance from the medium opposing the acceleration (the inertial force). Under deceleration, the profile reverses symmetrically.

Along a Lorentz-covariant, localized finite-energy branch satisfying the Laue condition ($\int T^i_i d^3x = 0$), the inertial coefficient is determined by the total Noether energy of the system [Noether, 1918, von Laue, 1911]. Coupling this same energy as the universal source of gravity yields the equivalence $m_i = m_g$ at leading order. The French satellite mission MICROSCOPE confirms the universality of free fall to a precision of 10^{-15} [Touboul et al., 2022].

For an oscillon in uniform, rectilinear motion, the wave pattern is transferred symmetrically across substrate nodes, leaving no dissipative wake or net energy loss. RefG identifies **frictionless inertial motion** with this coherent phase-relay regime: motion is the wave-like relay of form. Its limiting velocity is set by the maximum rate of phase transmission between nodes, from which the fundamental speed of light (c) emerges.

2.4 Local Unobservability and the Principle of Relativity

A local observer residing within a gravitational field reads clock rates and spatial dimensions as standard and unperturbed, because the observer’s clocks, measuring rods, and physical body are themselves constructed from the same foundation-medium. When substrate pressure shifts, the entire local physical system reconfigures synchronously. Relative to internal atomic transitions, slow variations in the gravitational field are strictly adiabatic: they preserve the vibrational mode and induce no spurious quantum transitions. Under this collective scaling, all dimensionless ratios—including the fine-structure constant α —remain absolutely invariant.

This ontological picture is constrained by comprehensive experimental bounds:

- **Gravitational Redshift:** Confirmed with exceptional precision from the classic Pound–Rebka experiment [Pound and Rebka, 1960] to the hydrogen maser on Gravity Probe A [Vessot et al., 1980] and recent eccentric-orbit tests with Galileo satellites [Delva et al., 2018, Herrmann et al., 2018], verifying the local dependence of clock rates on the gravitational potential.
- **Invariance of Dimensionless Ratios:** High-resolution quasar absorption spectra and laboratory optical atomic clocks constrain cosmological drifts in the fine-structure constant to $|\Delta\alpha/\alpha| \lesssim 10^{-6}$ [Kotuš et al., 2017]. RefG interprets this stability as consistent with local population-locked self-calibration.

The null result of the Michelson–Morley experiment reflects the medium’s **common-mode self-calibration**, whereas gravitational-wave detections by LIGO, Virgo, and KAGRA represent a **differential tensor response**:

- When the foundation-medium shifts uniformly and isotropically across a local domain, measuring rods, clocks, and optical propagation reconfigure synchronously in common mode, producing common-mode self-calibration in local measurements (modern optical cavities rule out spatial anisotropy in the speed of light to $\Delta c/c \lesssim 10^{-17}$; Herrmann et al. 2009).
- A gravitational wave, by contrast, is a time-dependent, quadrupolar differential strain ($h_+, h_\times$): it stretches space along one transverse axis while compressing it along the perpendicular axis. This differential deformation breaks common-mode cancellation, imprinting a measurable phase shift in interferometer arms.

FitzGerald–Lorentz length contraction [FitzGerald, 1889, Lorentz, 1892] is the historical precursor to this mechanism. RefG attributes this contraction to the fact that measuring instruments are themselves wave configurations of the medium.

In the local Minkowski limit, tensor waves propagate at the speed of light ($v_{\text{gw}} = c$). The simultaneous detection of gravitational waves and a gamma-ray burst from the binary neutron star merger GW170817 tightly constrains the fractional difference between gravitational and electromagnetic propagation speeds:

$$-3 \times 10^{-15} \leq \frac{v_{\text{gw}} - c}{c} \leq +7 \times 10^{-16} \quad (10)$$

[Abbott et al., 2017]. Concurrently, the orbital decay of relativistic binary pulsars (PSR B1913+16, PSR J0737-3039A/B) confirms general-relativistic quadrupole radiation damping to an accuracy of 10^{-3} – 10^{-4} [Weisberg and Huang, 2016, Kramer et al., 2021], while bounds on scalar breathing modes from the GWTC-3 catalog tightly constrain scalar wave channels [LIGO Scientific Collaboration et al., 2022].

The most definitive operational invariant of this framework is the proper time difference accumulated by two clocks traversing distinct gravitational and kinematic worldlines:

$$\Delta\tau = \int \sqrt{g_{\mu\nu} dx^\mu dx^\nu}. \quad (11)$$

The elapsed time displayed on reunited clock faces is an objective, absolute physical record of distinct pressure histories, whereas instantaneously measured local rates and sizes remain relational readouts within the common substrate.

2.5 Historical Context and the Role of RefG

The refractive, optical-medium reading of gravity possesses an illustrious heritage. In 1911–1912, prior to completing the geometric formulation of General Relativity, Albert Einstein related the coordinate speed of light to the gravitational potential via:

$$c(\Phi) = c_0 \left(1 + \frac{\Phi}{c^2} \right) \quad (12)$$

[Einstein, 1911, 1912]. Medium optics has even earlier roots: in 1851, Hippolyte Fizeau demonstrated that moving water partially drags light (governed by Fresnel’s dragging coefficient; Fizeau 1851), and in 1923, Walter Gordon proved that electromagnetic wave propagation in a moving dielectric medium obeys the null geodesics of an effective Riemannian metric [Gordon, 1923], establishing the first formal bridge between continuum optics and spacetime geometry.

In 1957, Robert Dicke formulated gravitation through a variable vacuum refractive index [Dicke, 1957]. The modern discipline of **analogue gravity** was inaugurated by William Unruh in 1981, demonstrating that sound waves in transsonic fluid flows propagate according to an effective acoustic metric exhibiting an event horizon [Unruh, 1981].

Grigory Volovik’s condensed-matter program [Volovik, 2003] and the extensive review by Barceló, Liberati, and Visser [Barceló et al., 2011] established how local Lorentz invariance and curved effective

geometries emerge from the microscopic dynamics of quantum condensates (such as superfluid Helium-3 and Helium-4). A notable milestone was achieved in 2019, when Jeff Steinhauer’s group experimentally observed the acoustic analogue of thermal Hawking radiation in a Rubidium Bose–Einstein condensate [Muñoz de Nova et al., 2019].

Parallel to the medium-optics approach stands the **thermodynamic and entropic** school of gravity:

- Andrei Sakharov interpreted gravity as an elastic vacuum phenomenon induced by quantum fluctuations [Sakharov, 1967];
- Ted Jacobson demonstrated that Einstein’s field equations arise directly as an equation of state from the thermodynamic entropy balance ($\delta Q = TdS$) across local Rindler horizons [Jacobson, 1995];
- Erik Verlinde formulated gravitational attraction as an entropic force driven by holographic information gradients [Verlinde, 2011].

These frameworks share with RefG the core conviction that spacetime geometry is emergent; in RefG, however, the effective geometry is generated not by abstract holographic screens, but by a concrete, pressure-refractive foundation-medium.

Refractive Gravity (RefG) introduces four fundamental ontological contributions to this historical continuum:

1. **Monistic Ontological Unity:** The foundation-medium is not merely a passive background for light; it generates matter itself—topological oscillons, measuring instruments, and observers. This moves beyond a passive-medium analogy and defines an independent, testable research framework.
2. **Local Population-Locked Cadence ($\Omega(x)$):** Nonlinear synchronization across the oscillon population renders local relativistic unobservability and the invariance of dimensionless constants an endogenous dynamical consequence of the system.
3. **The Universal Pressure-Deficit Channel:** RefG employs the pressure deficit ($\Delta P \sim -\rho_{\text{dyn}}$) as a unified language of source and readout, linking gravitational mass, time dilation, the factor-of-two light deflection, and tensor wave responses into a single energy balance.
4. **Topological Defect Strain and EFT Architecture:** The non-integrable strain imprinted by oscillon defects (strain incompatibility; Kleinert 2008) organizes at the macroscopic effective field theory (EFT) level into the Einstein–Hilbert action through Lovelock’s uniqueness result and Deser’s self-coupling construction [Lovelock, 1971, Deser, 1970], with the Einstein tensor $G_{\mu\nu}$ reflecting substrate defect density.

In this historical landscape, RefG unifies optical refraction, topological solitons, and thermodynamic equilibrium into an integrated research program, with every link subject to formal mathematical and observational verification.

3 Three Scales and Cosmological Horizons

3.1 The Local Scale: Compact Objects, Internal State, and External Readout

The universal pressure chain constructed throughout the preceding chapters manifests across three distinct astrophysical regimes:

1. **On the Local Scale:** As the localized pressure-deficit profile surrounding oscillons and compact relativistic bodies;

2. **On the Galactic Scale:** As the coherent, large-scale vortical flow and polarization of the foundation-medium;
3. **On the Cosmological Scale:** As the global expansion–relaxation of the foundation-medium and the coexistence of two descriptions of time.

These three phenomena are distinct dynamical regimes of a single foundation. We begin at the local scale, where this mechanism reaches its most extreme expression: compact relativistic objects.

In extreme gravitational regimes, an object’s **internal state** (M_{proper}) and its **externally read active gravitational mass** (M_{external}) decouple fundamentally. The RefG compact map strictly distinguishes the invariant internal energy reserve of accumulated matter from its externally read gravitational charge. An external observer measures the system through its pressure-deficit profile, clock dilation, and operational scale contraction, whereas the interior geometry, horizon formation, and global causal structure are determined by the complete nonlinear field equations.

Two-channel accounting separates an object’s local internal content from its asymptotic gravitational readout. The internal channel tracks local matter states, conserved topological charges, and proper volume; the external channel registers the deficit profile, clock retardation, operational dimensions, and total asymptotic mass. Gravitational compression sharply differentiates these two readouts without severing their common physical origin.

RefG attributes the mass defect of merging compact binaries to nonlinear reconfigurations in binding energy and the deficit profile. The quantitative connection to outgoing gravitational-wave flux is governed by the complete dynamical solution, benchmarked observationally by the binary black hole merger GW150914 [Abbott et al., 2016].

The biconformal metric governs the disparity between internal proper dimensions and external coordinate measures. On the compact branch of RefG, this divergence can become extreme: externally, such an object may appear ultra-compact, dark, and virtually “frozen,” while its internal proper volume remains vast. In both descriptions, the locally measured speed of light is invariant (c); what differs is the temporal and spatial readout constructed by a distant observer. The “frozen star” appearance is an external observational feature, while the historical intuition of a “dark star” dates back to John Michell in 1784 [Michell, 1784].

RefG interprets a **singularity** as a boundary of external readout: under intense deficit conditions, the externally read value of the internal scale approaches zero, while the local proper scale halts at a finite saturation limit, preventing internal collapse to zero volume. A unified description requires that local energy density and spacetime curvature remain strictly finite.

Roger Penrose’s singularity theorem [Penrose, 1965] relies on specific energy conditions, notably the Null Energy Condition (NEC: $T_{\mu\nu}k^\mu k^\nu \geq 0$). On the regular-core branch of RefG, the effective pressure source violates the classical NEC within the core zone via substrate elastic tension and confinement stresses. Because this energy condition is breached, Penrose’s theorem does not enforce a geometric singularity on this branch, admitting a **regular, non-singular interior core**.

Regular compact objects represent an active frontier in relativistic astrophysics. A prominent example is the Mazur–Mottola “gravastar”—a vacuum condensate object devoid of a singular core [Mazur and Mottola, 2004]. Such alternatives can be empirically distinguished from classical Kerr black holes via their Event Horizon Telescope (EHT) shadow morphology [Event Horizon Telescope Collaboration, 2019], quasinormal mode ringing, and late-time gravitational-wave “echoes” [Cardoso and Pani, 2019]. RefG shares structural kinship with this family, anchoring its regular interior in the pressure-energy balance of the universal foundation-medium.

3.2 The Galactic Scale: Vortical Flow and MOND

Observational Reality. In disk galaxies, rotation curves at large radii do not follow the Keplerian decline expected from visible baryonic matter alone: they exhibit rich morphological variety, frequently flattening or declining far more gently than Newtonian predictions. Concurrently, baryonic mass and asymptotic rotation speed obey a strict empirical law—the Baryonic Tully–Fisher Relation ($M_b \propto v_f^4$;

Tully and Fisher 1977, Lelli et al. 2016). This observational program began with Vera Rubin and Kent Ford’s measurement of the Andromeda galaxy [Rubin and Ford, 1970]. In the modern **Radial Acceleration Relation (RAR)**, 2,693 independent data points from 153 galaxies collapse onto a single universal curve [McGaugh et al., 2016].

The standard Λ CDM model accommodates these observations by postulating spherical dark matter halos. Modified Newtonian Dynamics (MOND), by contrast, introduces a universal acceleration threshold:

$$a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2, \quad (13)$$

below which gravitational acceleration transitions from Newtonian a_N to the deep-MOND regime $\sqrt{a_N a_0}$ [Milgrom, 1983]. Relativistic completions (such as TeVeS and Skordis–Złóšnik models) incorporate additional dynamical vector and scalar fields [Bekenstein, 2004, Skordis and Złóšnik, 2021].

The RefG Vortical Mechanism and Substrate Polarization. At galactic scales, RefG attributes the additional deficit to a large-scale **vortical flow and polarization** of the foundation-medium. An ordered rotational regime deepens the central pressure deficit, while baryonic oscillons and the background flow form a single, shared refractive profile.

At the macroscopic level, this response is described by a Ginzburg–Landau-type local effective potential:

$$W(g_v, g_N) = \frac{1}{3}g_v^3 + \frac{1}{2}g_N g_v^2 - a_0 g_N g_v, \quad (14)$$

where g_v is the foundation-medium response field, g_N is the baryonic Newtonian acceleration, and $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ is the universal transition acceleration scale. The variational equilibrium condition, $\partial W / \partial g_v = 0$, yields:

$$g \equiv g_N + g_v, \quad g_N = \frac{g^2}{g + a_0}. \quad (15)$$

In the deep asymptotic regime ($g \ll a_0$), this relation approaches $g^2 \simeq a_0 g_N$, yielding for circular orbits the Baryonic Tully–Fisher Relation ($v^4 = G M_b a_0$) without individual galaxy tuning parameters.

Observational Check on SPARC Data. Distinct from the published 153-galaxy, 2,693-point RAR sample, the cuts $\Delta V/V < 10\%$ and $V_{\text{obs}} > 20 \text{ km/s}$ select 2,788 points from 163 SPARC galaxies. With fixed a_0 and fixed mass-to-light ratios ($\Upsilon_{\text{disk}} = 0.5$, $\Upsilon_{\text{bulge}} = 0.7$), and without galaxy-by-galaxy parameter fitting, this comparison gives an RMS scatter of 0.141 dex and a mean residual of -0.012 dex (Figure 2).

Disk Geometry and QUMOND Numerical Test. For a finite-thickness Miyamoto–Nagai disk, transverse gradients of the algebraic field generate a geometric curl correction. On a 300×300 grid with $a = 3 \text{ kpc}$ and $b = 300 \text{ pc}$, Biot–Savart reconstruction gives, in this finite-grid configuration, a maximum radial-acceleration correction of 9.23% relative to the one-dimensional algebraic mapping over $a \leq r \leq 5a$.

Refractive Gravitational Lensing. Along the flat rotation curve branch, $\Phi_{\text{eff}} = v_f^2 \ln(r/r_0)$ and $n(r) \approx 1 - 2\Phi_{\text{eff}}/c^2$. Integrating the transverse refractive index gradient yields a deflection angle $\hat{\alpha} = 2\pi v_f^2/c^2$, matching the prediction of General Relativity for a singular isothermal sphere. Rotation and lensing thus emerge as two readouts of a single effective potential.

The closest modern analogue to this framework is the **Superfluid Dark Matter** model of Berezhiani and Khoury, where a MOND-like force arises from phonon excitations in a condensate [Berezhiani and Khoury, 2015]. However, while superfluid dark matter introduces an additional particle component, in RefG the coherent flow is a global regime of the unified foundation-medium itself.

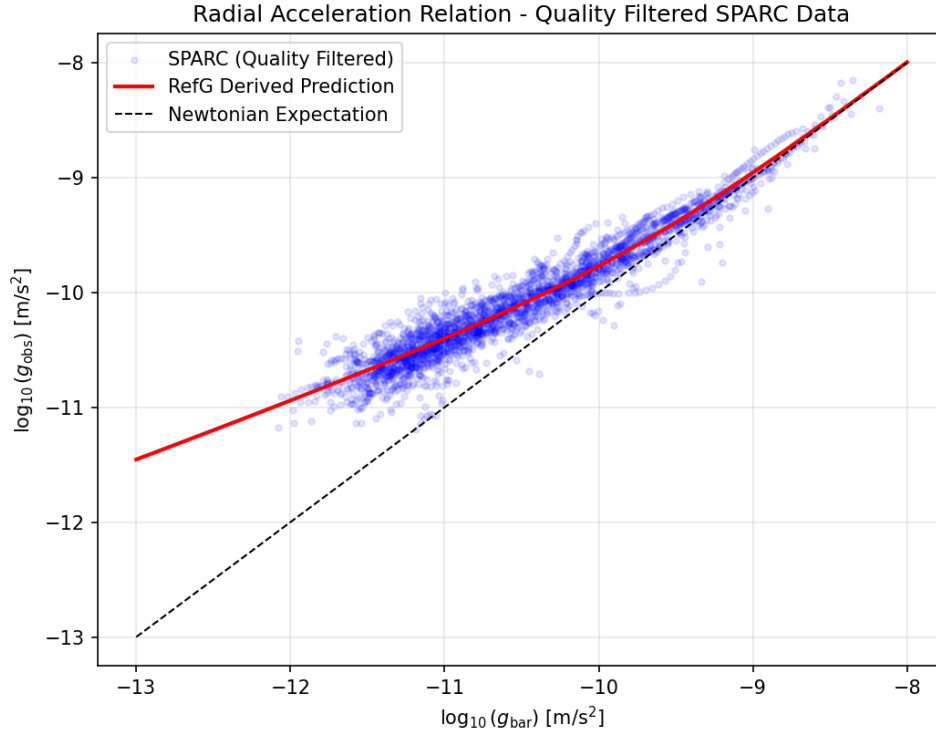


Figure 2: **Radial Acceleration Relation comparison with SPARC data.** The observed radial acceleration $g_{\text{obs}} = V_{\text{obs}}^2/R$ versus the baryonic acceleration g_{bar} for 2,788 selected points across 163 disk galaxies. With fixed a_0 and mass-to-light ratios, but no galaxy-specific fitting, the algebraic RefG curve obtained from $W(g_v, g_N)$ gives an RMS scatter of 0.141 dex and a mean residual of -0.012 dex.

Morphology. In RefG, galactic morphology reflects the flow regime of the foundation: ordered laminar rotation is associated with disk and spiral structure, whereas more chaotic or isotropic flow is associated with dispersion-supported elliptical galaxies.

Galaxy Clusters. In galaxy clusters, modifying the acceleration law alone does not fully resolve the dynamical mass deficit (historically highlighted by Fritz Zwicky in the Coma cluster; Zwicky 1933). RefG interprets the cluster mass budget through three distinct channels:

1. The global nodal deficit of the long-wavelength resonant network;
2. Individual galaxies and the long-range pressure-deficit fields of their constituent oscillons;
3. Cluster-scale vortical circulation and the hydrodynamic memory of past mergers.

All three enter a unified pressure-deficit budget, each counted exactly once.

Two decisive observational signatures distinguish this framework:

1. **Quantized Circulation:** If the substrate flow possesses a single-valued condensate phase, circulation is quantized in units of:

$$\Gamma = \oint \mathbf{v} \cdot d\mathbf{l} = n \frac{h}{m}, \quad (16)$$

where m is the effective mass of the flow mode. For ultralight modes ($m \sim 10^{-22}$ eV), vortex filaments could be detected via weak lensing or stellar disk kinematics [Hu et al., 2000, Hui et al., 2021].

2. **The Frozen Memory of the Bullet Cluster:** In the Bullet Cluster (1E 0657-558), weak lensing centroids are displaced from the X-ray gas at 8σ significance, tracking the collisionless galaxies [Clowe et al., 2006]. RefG interprets this separation as the **frozen memory of substrate vortical flow**: during collision, the gas was decelerated by hydrodynamic ram pressure, whereas the substrate deficit trace propagated forward with the galaxies. The memory timescale, spatial morphology, and lensing amplitude form a unified dynamical picture.

Thus, the unified observational program spans SPARC/RAR data [Lelli et al., 2016], cluster mass budgets, Bullet Cluster mergers, the CMB, BAO, and large-scale structure statistics.

3.3 The Cosmological Scale: Expansion of the Manifested Domain, Foundation Relaxation, and Two Descriptions of Time

The candidate beginning of the cosmological history of RefG is the centerless foundation-wide transition described in Section 1.2. It is read as the common boundary of the first distinguishable relations. Distance and direction acquire physical meaning only after those relations have formed. On the finite symmetric network examined in W3-39, where all vertices occupy equivalent states, a birth record strictly ordered by distance from one selected site breaks translation symmetry. This conditional obstruction does not apply to local waves or oscillon centers that arise later.

The candidate sequence is: the first relations; localization of oscillon seeds; overlap of resonant fields; formation of resonant coupling when the distinguishability threshold is crossed; network connection; formation of stable oscillons; and long-term expansion-relaxation. This ordering is a working hypothesis for cosmic genesis. The law that triggers the first transition, the initial partition of energy, the field equation for resonant propagation, the physical coupling threshold, and the selection of stable modes all remain open.

Two distinct pressures must be kept separate in this description. The foundation pressure P_F controls the cadence factor p and operational scaling. The thermodynamic pressure P_{th} is related to physical temperature through an equation of state, $P_{th}(\rho, T)$. A high value of P_F by itself determines

neither a high nor a low temperature. The initial thermal state must follow from the partition of energy among sectors, the equation of state for P_{th} , and energy exchange with radiation. The relation between T_{RefG} and the temperature reconstructed in standard cosmology is therefore open.

The observable physical content of the thermal state is carried by dimensionless ratios such as

$$\Theta_\gamma = \frac{k_B T_\gamma}{E_{\text{atom}}}, \quad \mathcal{R}_i = \frac{\Gamma_i}{H_\tau}. \quad (17)$$

Here E_{atom} is an atomic energy standard, Γ_i is the rate of the relevant microscopic process, and H_τ is the expansion rate measured in process time. A common p -rescaling can cancel from these ratios, so a fast early cadence alone cannot determine how hot the Universe was. A physical difference requires genuine energy transfer, an equation of state, or a measurable residue between the matter and photon channels. A joint derivation of the CMB spectrum, recombination, and BBN is a mandatory test of this thermodynamic map.

The quantity $R_{\text{act}}(t)$ denotes a post-origin local activation or correlation scale: within it, nodal relations at a given epoch reach the minimum distinguishability threshold ℓ_* . It is neither a radius of the Universe nor an external edge. On a homogeneous background, the same rule defines this local scale around every node comoving with the network. Its evolution law, the distinguishability threshold of the resonant field, and its relation to null directions for light in already activated space must be determined by the foundation field equation.

On the selected cosmological branch, formed matter follows an expanding nodal lattice of the foundation. The comoving separation of two galaxies in the mean flow is represented by N_{12} , the number of foundation units and links between them. In the ideal expansion component, N_{12} remains fixed while the foundation length of one link, $\ell_F(t)$, grows. Under that condition,

$$L_{12}(t) = N_{12} \ell_F(t). \quad (18)$$

The constancy of N_{12} applies only to the cosmological expansion component. Matter may move locally relative to the lattice, rotate, collide, or undergo gravitational displacement; in those processes N_{12} can change. In the mean Hubble flow, increasing separation is produced by the growth of the inter-node length scale.

The observer, their clock, and their ruler are built from the same foundation. The selected relaxation branch allows the material-standard scale p to contract as the mean foundation pressure decreases. Let

$$a(t) = \frac{\ell_F(t)}{\ell_{F,0}} \quad (19)$$

denote the foundation inter-node length scale and let $p(t)$ denote the material-standard scale. Observation reads their ratio,

$$A(t) = \frac{a(t)}{p(t)}. \quad (20)$$

The internal activation scale $R_{\text{act}}(t)$, the foundation length scale $a(t)$, and the material scale $p(t)$ are three distinct quantities. The first describes the local extent of stably distinguishable relations; the second describes changes in foundation link length; and the third describes the scaling of material size and cadence. The selected cosmological hypothesis relates them through common expansion–relaxation dynamics, whose quantitative law must follow from the foundation’s energy balance.

The common law proposed for local and global relaxation is a central hypothesis of RefG. In the candidate description of a strong-field compact object, low foundation pressure reduces the externally read size of matter through the p -scaling. For the mass of Earth, the Schwarzschild diameter is approximately 1.8 cm, comparable to the size of a hazelnut. The cosmological branch extends the same pressure–size–mass law to a homogeneous background: as pressure falls, the operational size of matter, its externally read mass, and its process cadence decrease. Here *external readout* means a measurement made from a region outside the object but still inside the Universe. The histories $P_F(t)$ and $a(t)$ remain to be derived. Our bodies, clocks, and rulers participate in the same change, so their contraction is not detected locally. In global observations, growth of the foundation length and contraction of material standards combine in the ratio $A = a/p$.

The Time Bridge and a Finite Process History. The selected pressure–cadence correspondence and common material scaling are defined by

$$p = \Omega = \sqrt{\frac{P_F}{P_{F,0}}}, \quad d\tau = p dt, \quad (21)$$

with $p_0 = 1$ fixing the normalization of the present background. Here p is the background process cadence of the foundation; in material clocks the same factor is realized as the local cadence Ω . Time t , counted by present-day clocks, lies on the finite interval $0 \leq t \leq T_0$, where the observational benchmark is $T_0 \approx 13.8$ Gyr [Planck Collaboration, 2020]. The postulated boundary of the first distinguishable relations is the common origin of both measures, $t = 0$ and $\tau = 0$. Process time is defined by

$$\tau(t) = \lim_{\epsilon \rightarrow 0^+} \int_{\epsilon}^t p(t') dt'. \quad (22)$$

On the branch selected in this monograph, finiteness of process history is imposed as an initial condition:

$$\tau_0 = \lim_{\epsilon \rightarrow 0^+} \int_{\epsilon}^{T_0} p(t') dt' < \infty. \quad (23)$$

The early value of p may be extremely large while remaining integrable. If $p \geq 1$ almost everywhere on $(0, T_0]$ and $p > 1$ on a set of nonzero measure, then $\tau_0 > T_0$: process history is finite but contains more internal cycles than the interval counted by present-day clocks. The exact value of τ_0 must be derived from the dynamics of $P_F(t)$ and is not fixed in advance. The history of each material structure begins at its own formation epoch and retains the process cycles accumulated thereafter.

Time Dilation of Supernovae. On the selected branch, the larger early value of p makes material processes faster relative to today’s cadence. Recovering the observed supernova time dilation additionally requires a complete map of photon propagation and atomic standards. If that map identifies spectroscopic redshift with the operational ratio, then

$$\frac{A_0}{A_e} = \frac{a_0/p_0}{a_e/p_e} = 1 + z. \quad (24)$$

An event lasting $\Delta\tau_e$ in the process time of the source is then conditionally recorded by the present observer with duration

$$\Delta\tau_{\text{obs}} = \frac{A_0}{A_e} \Delta\tau_e = (1 + z) \Delta\tau_e. \quad (25)$$

Type Ia supernova light curves and spectral aging exhibit this $(1 + z)$ time dilation [Blondin et al., 2008]. Standard cosmology attributes the law to metric expansion. The candidate RefG account relates it to expansion of the foundation scale a , the material cadence p , and the p^2 propagation channel of light; the full photon–atomic derivation remains to be completed.

Early Structural Maturity (JWST). A supernova time profile compares clocks at different epochs through photon propagation. The maturity of a galaxy records accumulated process history: mass assembly, successive stellar generations, chemical enrichment, and dynamical organization all require many internal cycles. A larger early p may provide more accumulated process time. This extra time does not by itself prove galaxy growth; an independent model of collapse, cooling, star formation, and feedback is required. The early massive galaxies observed by JWST are therefore a stringent test of this branch [Labbé et al., 2023].

Two Observational Channels for One Mechanism. Supernova time dilation and the maturity of early galaxies constitute two distinct tests. The first probes the photon–clock map; the second probes whether accumulated process time produces real structural effects. Passing both tests would support the common picture of global foundation relaxation and the historical slowing of time.

Ontological Consistency and the Arrow of Time. Within a single-medium ontology, the familiar **vacuum-energy problem**—in which the sum of quantum zero-point fluctuations exceeds the observed scale associated with Λ by approximately 120 orders of magnitude—takes a different accounting form: every channel must enter the common energy balance of one medium. In the candidate reading of RefG, the small cosmic acceleration may reflect a residual dynamical departure from that balance. Related mechanisms have been developed in the q -theory program of Volovik and Klinkhamer [Volovik, 2003, Klinkhamer and Volovik, 2008].

A working hypothesis of RefG relates the macroscopic **arrow of time** to the global direction of foundation relaxation. Like a colored crystal dissolving in water, energy dispersed through the medium does not spontaneously reconcentrate into a solid localized crystal. Such statistical irreversibility [Planck, 1899] may connect the thermodynamic, cosmological, and radiative arrows to one source. A complete derivation of entropy production and irreversible dynamics remains open.

3.4 Quantum Randomness and Two-Level Time

Two-level time provides a clarifying lens for quantum phenomena. The spacetime coordinates of a measurement event are recorded in oscillon (clock) time, while the selection of a specific outcome is associated with the deeper, pre-clock state of the foundation. This pre-clock level cannot be measured directly by clocks, because clocks are themselves emergent oscillon systems.

Bell’s theorem and its experimental verifications rule out local hidden-variable theories [Bell, 1964, Hensen et al., 2015]. Meanwhile, the **no-signaling theorem** ensures that entanglement correlations cannot transmit superluminal messages.

RefG interprets this as **atemporal internal coordination**: within the internal language of the foundation, entanglement is not a physical signal propagating through space; in the metric language of clocks, it appears as an instantaneous statistical correlation between separated measurement outcomes.

Wheeler’s **delayed-choice experiment** is understood in the same terms [Wheeler, 1978, Jacques et al., 2007]: a delayed measurement choice determines whether wave interference or which-way particle information is realized. RefG’s two-level time describes this without retrocausal signaling, as an expression of the underlying phase integrity of the state.

Related concepts in modern physics include:

- The atemporal ground level of the **Wheeler–DeWitt equation** ($\hat{H}\Psi = 0$; DeWitt 1967);
- The **Page–Wootters mechanism**, where time emerges from subsystem quantum correlations [Page and Wootters, 1983];
- **de Broglie–Bohm pilot-wave theory**, characterized by non-local phase coordination [Bohm, 1952].

This quantum framework is governed by two strict quantitative requirements:

1. **Born’s Probability Rule** ($P = |\psi|^2$): The model must derive the observed statistical distribution of measurement outcomes [Born, 1926];
2. **The No-Signaling Condition**: Internal phase coordination must never generate a transmissible signal that violates relativistic causality.

4 Oscillons and the Origin of Matter

4.1 Many Patterns in One Ocean

To build physical intuition, imagine a single continuous ocean and the diverse wave motions that arise within it. Most waves form, propagate across the surface, and quickly dissipate; under specific nonlinear conditions, however, certain configurations self-organize into stable, localized, self-sustaining structures:

- One may form a toroidal vortex ring, with fluid circulating continuously in a closed loop;
- Another may manifest as a stable spherical pulsation (soliton);
- A third may take shape as a helical, twisting wavepacket.

This hydrodynamic picture illustrates the foundational thesis of this chapter: **Electrons, protons, photons, and quarks are distinct resonant and topological patterns supported by a single universal foundation-medium.** The continuous medium itself oscillates: massive particles are locally confined modes of this oscillation, whereas photons are freely propagating transverse wave states.

A historical precursor is Tony Skyrme’s model, where nucleons (protons and neutrons) are described as topological solitons—“skyrmions”—of a nonlinear meson field [Skyrme, 1961]. RefG elevates this intuition to the fundamental ontological level: all such soliton-like structures are normal modes of a single universal foundation-medium.

The classical image of a rigid, pointlike particle gives way to a **continuous wave-medium ontology**: spatial extent is characterized by the vibrational amplitude and phase of the foundation, while a localized event registered in a detector represents the resonant coupling between this extended vibration and the internal oscillon modes of the detector’s atoms. RefG thus grounds the spatial amplitude and phase of the quantum wave function in an explicit ontological substrate.

In an atom, the “electron cloud” is an actual vibrational amplitude distribution rather than the rapid orbital transit of a point mass. The extended state maintains a coherent phase structure throughout space, and the localized point of measurement emerges through local resonant locking with the detector.

The single-electron double-slit experiment synthesizes two foundational realities: the wave amplitude depends simultaneously on the global geometry of both slits, yet each detection event on the screen manifests as a discrete, localized point. Akira Tonomura’s group directly recorded this progressive statistical accumulation using electron microscopy [Tonomura et al., 1989]. RefG interprets an atom’s stationary electronic state as a standing, confined wave mode of the medium rather than an accelerating classical point charge; consequently, electromagnetic radiation occurs only during transitions between discrete states.

The spatial spread of a wavepacket and the intrinsic structural form factor of a particle are distinct physical concepts:

- Elastic scattering probes charge and energy distribution form factors, while deep inelastic scattering reveals internal parton structure and distribution functions;
- The spatial extent of a propagating wavepacket does not enter these observables as an intrinsic physical size.

Empirically, the electron exhibits a pointlike electromagnetic form factor down to scales of order 10^{-18} m [Navas et al., 2024]. A direct empirical test of the oscillon framework is whether the form factor computed from its internal topological charge profile reproduces this pointlike behavior across accessible momentum transfer scales.

4.2 Localized Confinement: Massive and Massless Modes

In the RefG particle framework, the defining ontological dividing line between massive and massless sectors is **localized confinement**.

Massive Oscillons. The energy of a massive particle is confined within a localized, self-sustaining configuration. When a wavepacket is trapped via vortex circulation or stable nonlinear pulsation within a finite spatial region, Bernoulli-type dynamics induce a persistent static pressure deficit in the surrounding medium.

Long-lived localized nonlinear pulsations are well-known in classical field theory as **pulsions and oscillons** [Bogolyubsky and Makhankov, 1976, Gleiser, 1994, Copeland et al., 1995]. RefG proposes

that their stability arises from population-level resonant locking and conserved topological charges. In external readout, this locally confined energy manifests as inertia and effective gravitational mass.

While standard field-theoretic oscillons eventually decay through radiation leakage, RefG proposes nonlinear phase-locking with the surrounding particle population as the mechanism that suppresses radiative loss and selects long-lived particle states.

Massless Carriers (The Photon). A massless mode possesses neither a rest frame nor a localized static core—it propagates through the network at the limiting phase transmission velocity (c). RefG interprets the photon as a freely propagating transverse phase mode.

Helicity and localized rotation are fundamentally distinct phenomena:

- The ± 1 helicity of a photon describes the spatial spirality (handedness) of transverse electric and magnetic phase vectors along the propagation axis, without forming a localized static vortex;
- A massive mode possesses a localized topological core and a static pressure-deficit well.

The energy-momentum tensor $T_{\mu\nu}$ of a photon couples fully to the gravitational energy balance: while free electromagnetic radiation does not create a static, localized deficit well like a massive oscillon, its propagating energy enters the total gravitational source of the system.

The same principle applies to composite hadrons: inside protons and neutrons, the confined energy of strong quark and gluon interactions does not register as a separate external field, but contributes fully to the hadron’s total inertial energy and externally read gravitational mass.

Local topological form, spin, electric charge, color, and polarization preserve internal quantum distinctions, while gravitational readout unifies them through a single pressure-deficit channel.

4.3 Self-Forming Resonators and the Two-Level Filter

Nonlinear wave equations admit an infinite diversity of mathematical solutions; nature, however, realizes a sparse, strictly defined spectrum of stable elementary particles. Why does nature select this specific discrete set?

RefG organizes this proposed selection mechanism through a **two-level dynamical filter**:

1. Level One: Kinematic and Population Filter. An acoustic handpan provides an intuitive analogy. While the resonant length of a guitar string is externally fixed by its wooden body, a three-dimensional oscillon must sculpt its own resonant cavity within the medium. On a handpan’s surface, a stable acoustic note unites the fundamental frequency, its octave, and the compound fifth into a harmonic 1 : 2 : 3 triad.

This triad is a structural rhythm: a self-forming resonator binds several mutually coherent modes into a single physical entity. In RefG, an oscillon’s pressure deficit alters the local geometry, which in turn filters viable modes: a stable mode must propagate through the substrate while locking into resonant coexistence with the surrounding particle population.

2. Level Two: Energetic Stability Filter. Any mode that passes the kinematic filter but possesses an energetically accessible decay channel rapidly disintegrates into lighter, stable ground states.

Hadronic resonances (Δ, ρ, ω) illustrate this: with typical lifetimes of 10^{-22} – 10^{-24} s, they appear in scattering spectra but cannot endure as asymptotic particles, shedding energy into stable channels (pions, protons).

Electron decay has never been observed; as the lightest charged lepton, it is protected by electric charge conservation. The extraordinary experimental stability of the proton ($\tau_p > 10^{34}$ years) is linked in RefG to corresponding topological protection.

The mathematical language governing these filters is that of nonlinear **attractors**—stable phase-space configurations toward which dynamical trajectories asymptotically converge.

A stable particle represents a self-consistent equilibrium: the wave configuration carves a pressure deficit in the medium, and this deficit acts as a potential well that stabilizes and reinforces the configuration. In classical field theory, Sidney Coleman’s Q-balls are protected by conserved internal Noether charges [Coleman, 1985]. In RefG, stability is reinforced by nonlinear phase-locking across the entire population.

This collective synchronization requires no external coordinator: Christiaan Huygens’s classic observation of pendulum clocks synchronizing through a common wooden beam [Huygens, 1665], mode-locking in lasers, and Yoshiki Kuramoto’s nonlinear synchronization model [Kuramoto, 1975] demonstrate how collective order emerges endogenously. In particle physics, this vision resonates with Geoffrey Chew’s S-matrix “Bootstrap” program, where particle spectra form a self-consistent whole [Chew and Frautschi, 1961].

Under adiabatic gravitational changes, resonant locks are preserved: the collective cadence $\Omega(x)$, operational scale, and externally read mass adjust synchronously, while dimensionless ratios and phase harmonies remain invariant. Rapid, high-energy shocks can break this locking, redistributing energy into other compatible channels.

4.4 The Global Resonator: Unity of Micro- and Macro-Spectra

The two-level filter specifies the proposed selection mechanism; we now ask where it is realized.

Ernst Chladni’s classic acoustic experiment provides the guiding intuition [Chladni, 1787]: sand scattered across a vibrating metal plate collects along the nodal lines of standing waves, where the oscillation amplitude is minimal. In the RefG map, this image corresponds to global modes and their nodes. Every resonator possesses its own spectrum: its geometry and boundary conditions determine which modes are amplified and which are suppressed.

After the operational network has formed, RefG treats it as a candidate **unified global volumetric resonator**. Only at this stage does the concept of an eigenspectrum acquire a defined meaning: the connection operator, network geometry, and boundary conditions select the admissible modes. The three-dimensionality of this resonator and the identification of its spectrum with physical particles and the cosmic web remain central hypotheses. During subsequent expansion and thermal evolution, its modes may lock into stable physical forms:

- **Short-Wavelength Modes** at high frequencies correspond to localized elementary particles, or oscillons;
- **Long-Wavelength Modes** at low frequencies correspond to the large-scale distribution of matter, the **Cosmic Web**.

According to the central hypothesis, the distribution of galaxies and galaxy clusters may resemble sand on a Chladni plate: filaments and vast cosmic voids may reflect the nodal pattern of standing modes in the global resonator. This identification requires a dedicated spectral test.

Standard cosmology describes this large-scale morphology through Yakov Zel’dovich’s anisotropic gravitational collapse [Zel’dovich, 1970]. RefG relates the same morphology to the ontological language of a single spectral operator.

The central hypothesis of this map is that **the microscopic elementary particle and the macroscopic cosmic web are two different readouts of one universal spectral operator**: one is a locally confined topological form, and the other is a globally extended cosmic pattern.

A joint empirical test of this operator must encompass:

1. The localized particle mass spectrum;
2. The cosmological history of structure formation;
3. The mass budgets of galaxy clusters;

4. The Baryon Acoustic Oscillation standard ruler at approximately $100 h^{-1}$ Mpc [Eisenstein et al., 2005].

Within the candidate picture of a finite, boundaryless resonator, conceptually related to Christof Wetterich’s variable-gravity framework [Wetterich, 2013], cosmological expansion describes the evolving relation between the geometry of the foundation and its own internal material standards. No external embedding container is required.

In current CMB observations, the low-multipole anomalies at $l = 2, 3$ remain moderate. Searches for cosmic topology establish topology-dependent lower bounds and do not select a unique global form [Planck Collaboration, 2016].

4.5 Discreteness and the C_3 Generation Symmetry

In the Standard Model, the masses of the charged leptons—electron (e), muon (μ), and tau (τ)—are arbitrary Yukawa coupling inputs. RefG treats them as the **ninth-order internal three-phase (C_3) triad** of a single topological defect: three distinct resonant modes of one underlying physical structure.

The charged-lepton mass parametrization utilizes the vibrational scaling $m \propto \nu^2$ (or $\nu \propto \sqrt{m}$) within a cyclic three-phase geometry:

$$\nu_k = \mu \left[1 + A_K \cos \left(\theta + \frac{2\pi k}{3} \right) \right], \quad k = 0, 1, 2, \quad (26)$$

where:

- μ is an overall frequency scale;
- A_K is the dimensionless three-phase modulation amplitude;
- θ is the mass phase angle;
- $k = 0, 1, 2$ indexes the three generations (e, μ, τ).

When $A_K = \sqrt{2}$, the C_3 algebra identically recovers Yoshio Koide’s empirical mass relation [Koide, 1982, 1983]:

$$K = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \quad (27)$$

This three-phase parametrization possesses a rich literature:

- Carl Brannen formulated the relation in terms of circulant matrix eigenspaces [Brannen, 2006];
- Robert Foot demonstrated that $K = 2/3$ corresponds to a 45° geometric orientation in $\sqrt{m}$ -space [Foot, 1994];
- Brannen’s Pancharatnam–Berry phase construction yielded $\pi/12$ while leaving $\theta = 2/9$ as a dynamical parameter [Brannen, 2010];
- Konstantin Shulga obtained $\theta_\ell = -2/9$ within a compact family cycle [Shulga, 2026].

RefG offers a candidate route connecting this structure to a **ninth-order holonomic return cycle**:

1. On the selected $h = 2$ branch, the first non-trivial topological return gives the candidate mass angle:

$$\theta = \frac{h}{9} = \frac{2}{9} \text{ rad.} \quad (28)$$

2. $A_K = \sqrt{2}$ represents energetic equipartition between internal singlet and doublet vibrational channels.

A simple C_3 -invariant potential exhibits **θ -blindness**—an inability to select θ dynamically—showing that a simple potential minimum is insufficient and motivating closed topological holonomy as a candidate origin of the mass angle.

Evaluating $(A_K, \theta) = (\sqrt{2}, 2/9)$ at tree level, anchored to the physical electron pole mass, yields deviations of 451.7σ for the muon mass and 0.61σ for the tau mass. Reconciling this tree-level geometry with measured pole masses requires an action-derived radiative correction bridge [Sumino, 2009].

This structure has a clear acoustic analogue: **The C_3 /Koide triad of charged leptons is an anharmonic, stable chord supported by a 3D volumetric resonator**; within the candidate map, all three generations are normal modes of a single topological defect.

5 Topology and Quantum Numbers

5.1 Phase, Orientation, and Topological Quantization

Thus far, the foundation-medium has been described primarily in terms of vibrational amplitude and local pressure. However, the complete local state of the medium is characterized by multiple coupled field variables: the phase-clock scalar, amplitude profile, and spatial orientation frame each possess distinct degrees of freedom. In this monograph, Φ represents the compact designation for the complete state of the foundation-medium rather than a simple single-component scalar: its full configuration simultaneously carries amplitude, phase, internal geometric form, and spatial orientation.

If Φ were merely a single-valued scalar field, its velocity field would be defined strictly by the gradient $\mathbf{u} = \nabla\Phi$, forcing the curl (vorticity) to vanish identically ($\nabla \times \mathbf{u} = 0$). The emergence of a non-trivial topological sector requires:

- **A multi-valued phase,**
- **A singular defect nodal line,** and
- **A localized internal spatial orientation frame.**

At the central core of a defect line, the wave amplitude vanishes ($|\psi| = 0$), while around it the phase winds by an integer multiple of 2π ($\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n$). A canonical laboratory analogue is a quantized vortex filament in superfluid Helium.

The cosmological analogue is a **cosmic string**—a macroscopic topological defect line formed during a symmetry-breaking phase transition in the early Universe via the Kibble mechanism [Kibble, 1976]. In RefG, this represents a “frozen seam” of the foundation: vacuum topology determines whether such lines form, while astrophysical observations (CMB lensing and pulsar timing arrays) tightly constrain their cosmological tension.

Lord Kelvin’s seminal papers on vortex atoms and vortex motion [Thomson, 1867, 1869] inspired Peter Guthrie Tait to systematically catalog knots, laying the foundations of modern mathematical knot theory [Tait, 1877]. RefG continues this lineage using the modern vocabulary of pressure-energy balance and population-locked resonance.

History provides a crucial lesson: 19th-century vortex atoms in ideal inviscid fluids failed to yield a stable discrete spectrum or reproduce atomic spectra. In RefG, the topological knot provides the kinematic scaffolding of form, while physical viability and dynamical stability are secured by pressure-energy feedback and nonlinear phase-locking across the oscillon population.

Phase quantization occurs when the field is single-valued and continuous along a closed loop, returning to its initial state modulo 2π , with the loop belonging to a non-trivial first homotopy group ($\pi_1 \neq 0$).

For a closed, orientable, framed ribbon, the **Călugăreanu–White–Fuller (CWF) theorem** applies [Călugăreanu, 1961, White, 1969, Fuller, 1971, Ricca and Nipoti, 2011]:

$$Lk = Wr + Tw, \tag{29}$$

where:

- Lk (Linking Number) is the topological integer invariant measuring total linking;
- Wr (Writhe) measures the 3D spatial coiling of the ribbon’s central axis;
- Tw (Twist) measures the internal phase winding of the ribbon frame around its axis.

Under continuous deformation, Wr and Tw can vary continuously (e.g., spatial coiling converting into internal twist), yet their algebraic sum Lk remains strictly invariant.

In fluid dynamics, kinetic helicity ($H = \int \mathbf{u} \cdot (\nabla \times \mathbf{u}) d^3x$) is an invariant of ideal barotropic flow, which Keith Moffatt connected to the topological linking number Lk [Moffatt, 1969]. In superfluid Helium, circulation quantization was predicted by Lars Onsager and Richard Feynman [Onsager, 1949, Feynman, 1955] and experimentally verified by Joe Vinen [Vinen, 1961]. Integer phase winding is thus an established laboratory reality.

The core principle “**Topological Confinement = Conserved Charge**” implies in RefG that electric charge is the external readout of a closed phase-orientation winding: topological invariants define the discrete charge class, while field dynamics govern its electromagnetic coupling.

In the RefG synthesis, **spin-1/2 and elementary electric charge** are read as two candidate geometric traces of a common double-covered topological source: the half-twist determines spinorial transformation under rotation, while chiral orientation and integer winding label the charge class and sign. Their quantitative completion requires separate derivations of the spin transformation law and electric-flux normalization.

Direct experimental analogues exist: Kelvin waves on quantized vortices in superfluid Helium have been directly excited and their dispersion measured [Minowa et al., 2025]; in classical water vortices, helical axial modes and their dispersion relations have been mapped experimentally [Barckicke et al., 2026]. In RefG, an oscillon’s radial breathing mode corresponds to its Compton mass pulsation, while its helicoidal mode governs charge and radiative channels.

5.2 Spin and Fermionic Statistics

A phase field twisted by a half-turn (π) along an oscillon’s internal axis generates a **Möbius-strip-type double-covered Z_2 topology**:

- A single spatial rotation of 2π reverses frame orientation, flipping the sign of the wave amplitude ($\psi \rightarrow -\psi$);
- A double rotation of 4π fully restores the initial quantum state ($\psi \rightarrow +\psi$).

This Z_2 geometry embodies the fundamental transformation property of the **spinor representation** ($SU(2)$). RefG associates electron spin-1/2 with this internal double-covering. Changing this topological class requires tearing the configuration: vanishing amplitude at the core, defect reconnection, or altering boundary conditions.

Completing the connection to Fermi–Dirac statistics requires that exchanging two identical oscillons produces a phase shift of π , yielding an antisymmetric multi-particle state in Fock space:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1). \quad (30)$$

Under the required exchange antisymmetry, the amplitude for two identical fermions occupying the same state vanishes identically ($\Psi = -\Psi \implies \Psi = 0$). Deriving this exchange phase from the Möbius topology is the remaining step needed to recover the **Pauli Exclusion Principle**. For bosonic modes, the Möbius half-twist is absent, the state remains symmetric under exchange ($\Psi \rightarrow +\Psi$), and macroscopic Bose condensation is permitted.

In relativistic quantum field theory, the Spin-Statistics Theorem unites Lorentz invariance, positive energy, and microcausality [Pauli, 1940]. RefG’s topological framework offers a candidate geometric mechanism for this connection.

5.3 Charge, the Photon, and the Electromagnetic Channel

RefG defines **electric charge** as the external spatial trace of an oscillon's internal chiral phase twist:

- Right-handed chiral twist corresponds to positive charge ($+e$);
- Left-handed chiral twist corresponds to negative charge ($-e$).

Externally, this trace extends radially in all directions. In quantitative electrodynamics, its surface integral must satisfy Gauss's Law:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0}. \quad (31)$$

A **positron** is the exact topological mirror image of an electron (possessing opposite chiral twist). In electron-positron annihilation ($e^+ + e^- \rightarrow 2\gamma$), opposite topological windings untwist and cancel, releasing locally confined energy into freely propagating transverse radiation (photons). In a neutral oscillon, opposing chiral twists cancel internally, yielding zero net flux (characterizing the neutrino sector).

A uniform rotation of the internal phase angle corresponds to global $U(1)$ gauge symmetry. Describing spacetime-dependent rotations necessitates a local frame connection and covariant derivative. RefG identifies the **photon** as a **transverse, helicoidal wave** of this connection: it propagates phase-twist disturbances across space, acting as the massless gauge boson of electromagnetism.

The topological nature of electromagnetism is manifest in classical electrodynamics: Antonio Rañada's vacuum solutions exhibit linked, knotted electric and magnetic field lines classified by the Hopf invariant [Rañada, 1989].

The sign of physical forces (attraction vs. repulsion) is governed by gauge boson spin, source type, and coupling structure:

- **Vector Gauge Channel (Spin-1):** Like charges repel, opposite charges attract;
- **Tensor Gravitational Channel (Spin-2):** All positive energy densities attract universally.

In three spatial dimensions, a conserved massless flux dilutes as $1/r^2$, yielding a $1/r$ potential.

Hydrodynamic Bjerknes forces [Bjerknes, 1906] illustrate the limits of simple fluid analogies: in an acoustic medium, bodies pulsating in-phase attract, while out-of-phase bodies repel—the exact opposite of electrostatics. This demonstrates that force signs are not reducible to crude mechanical collisions; they are governed by the rigorous mathematics of vector gauge fields, source currents, and topological linking.

The fine-structure constant $\alpha \approx 1/137$ sets the dimensionless coupling strength of this electromagnetic channel and is interpreted as the impedance ratio of the substrate's electric-flux and transverse-photon response channels.

5.4 Color Confinement and Three Generations

The three-dimensionality of space ($D = 3$) plays a crucial structural role in the RefG particle map: **The three-axis spatial orientation frame of an oscillon provides the geometric substrate for the color charge triad (red, green, blue: r, g, b).** Color represents the internal spatial degree of freedom of this frame:

- **Quark:** An oscillon mode vibrating along a single internal axis is topologically open, possessing unconfined endpoints (an open topological segment);
- **Baryon (Proton, Neutron):** The symmetric combination of all three axes ($r + g + b$) forms a fully closed, stable 3D topological knot (a color-singlet white state);
- **Meson:** Pairing an axis with its opposite anti-axis (color + anticolor) likewise forms a closed, color-neutral loop.

RefG identifies the candidate mechanism for **color confinement in Quantum Chromodynamics (QCD)** with the geometric incompleteness of an isolated quark: a closed, gauge-invariant state cannot be formed from a single open axis.

The mode count of an unconstrained three-axis deformable oscillon matches the 8 generators of $SU(3)$, establishing the structural map of the color sector. Formal identity requires verifying the $SU(3)$ structure constants, local gauge invariance, and interaction vertices.

The **Three-Generation Puzzle** is a proposed extension of the C_3 structure explored in Chapter IV:

- Charged leptons (e, μ, τ) represent the ninth-order three-phase vibrational modes of a single topological defect;
- Quark generations ($u/d, c/s, t/b$) replicate this generational architecture across the three-axis colored sector.

In this candidate map, the three generations are read as successive holonomic return modes of a single 3D volumetric resonator.

The universal foundation of holonomy is **Berry's geometric phase** [Berry, 1984]: under adiabatic cyclic transport across parameter space, a quantum state acquires an invariant geometric holonomy phase alongside its dynamical phase. RefG links this to the ninth-order return cycle of a framed defect, yielding $\theta = 2/9$ rad on the $h = 2$ branch.

Why is Space Three-Dimensional? In RefG, $D = 3$ is intimately tied to the **stability of topological knots**:

- Closed one-dimensional loops form stable, non-trivial knots and links exclusively in $D = 3$ spatial dimensions;
- In higher dimensions ($D \geq 4$), any knotted loop can be untied and unlinked smoothly along extra spatial dimensions without self-intersection.

This topological argument builds upon the insights of Paul Ehrenfest, Gerald Whitrow, and modern investigators [Ehrenfest, 1918, Whitrow, 1955, Berera et al., 2017]. RefG formulates the co-emergence of material discreteness and 3D space from a single topological locking mechanism as a central working hypothesis.

5.5 Geometric Map of the Standard Model

RefG organizes the particle spectrum of the Standard Model into the following geometric architecture:

1. The Electron and Charged Leptons. The electron is a complete toroidal, framed Möbius Z_2 topological defect with all three spatial axes locally closed. RefG connects this double-covered geometry to the Dirac factor $g = 2$.

The electron's measured magnetic moment slightly exceeds 2. Julian Schwinger calculated the leading quantum electrodynamic correction ($\alpha/2\pi$), while modern Penning trap experiments measure $|g_e|/2 = 1.00115965218059(13)$ [Schwinger, 1948, Fan et al., 2023]. This precision measurement provides a benchmark for testing the quantum completion of the topological map.

2. Quarks and Color Charge. Quarks are open axial topological segments. Their fractional electric charges ($+2/3, -1/3$) are associated with one-third subdivisions of the 3D closed loop; their exact values remain to be fixed by canonical charge normalization. Color confinement dictates that only closed, color-singlet configurations (baryons and mesons) exist as asymptotic states.

3. Gauge Bosons.

- **Photon:** Massless propagating transverse helicoidal wave ($U(1)$ gauge symmetry, 2 helicities);
- **Gluons:** 8 massless transverse modes mediating color transitions between internal axes ($SU(3)$ octet);
- **$W^\pm$ and Z^0 Bosons:** Locally confined, massive carrier oscillons mediating electroweak transitions ($SU(2)_L \times U(1)_Y$).

RefG interprets **beta decay** ($n \rightarrow p + e^- + \bar{\nu}_e$) as a topological reconnection: altering the internal axis orientation ($d \rightarrow u$) produces a transient charged weak carrier (W^-) that decays into leptonic channels, bringing chirality, energy–momentum balance, and the decay spectrum into a single geometric transition.

4. Neutrinos and the Higgs Boson.

- **Neutrinos:** Electrically neutral, incompletely confined chiral phase modes. Their incomplete topological confinement is associated with their ultra-small masses ($m_\nu < 0.1$ eV) and weak coupling, while Pontecorvo flavor oscillations represent phase interference among mass eigenspaces;
- **Higgs Boson:** A purely spherical, radial breathing pulsation of the foundation-medium devoid of angular momentum, naturally corresponding to a spin-0 scalar state ($m_H \approx 125.25$ GeV).

5. The Graviton and the Weinberg–Witten Theorem. The tensor wave sector contains two transverse-traceless polarizations ($h_+, h_\times$). The Weinberg–Witten theorem [Weinberg and Witten, 1980] forbids massless spin-2 particles with a Lorentz-covariant, gauge-invariant energy-momentum tensor in flat spacetime. At the low-energy effective level, RefG can meet this constraint only if the same branch simultaneously carries a massless tensor mode, universal coupling, and effective Lorentz symmetry. The graviton is then read not as an isolated fundamental particle on a fixed background, but as a collective tensor-shear excitation of the foundation-medium.

6. Nonlinear Soliton Lagrangian. The formal mathematical foundation of this topological map relies on nonlinear field Lagrangians admitting stable knotted soliton solutions (in the spirit of the Faddeev–Niemi model; Faddeev and Niemi 1997).

5.6 Fundamental Constants: α and the Gravitational Hierarchy

The fine-structure constant:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.035999} \quad (32)$$

remains one of the deepest mysteries in physics, described by Richard Feynman as one of the greatest unexplained numbers in science [Feynman, 1985].

In RefG, α is interpreted as the **dimensionless impedance ratio of the substrate’s electric-flux and transverse-photon response channels**.

This definition is governed by three precise distinctions:

1. Simple ratios of Planck and electron masses or frequencies do not yield 137;
2. The identity $r_e/\bar{\lambda}_c = \alpha$ is an algebraic consequence of dimensional definitions;
3. Integer phase-locking steps represent physical realities (as demonstrated by the Josephson effect and Shapiro steps; Josephson 1962, Shapiro 1963), though seconds are themselves calibrated by atomic oscillons.

Thus, α is not an absolute frequency or energy; it is the **dimensionless impedance ratio of the substrate’s electric-flux and transverse-photon response channels**. Under local pressure shifts, both responses scale synchronously, leaving α invariant.

The Hierarchy Problem: Why is Gravity So Weak? The electrostatic repulsion between two protons exceeds their gravitational attraction by a factor of 10^{36} , while for electrons this ratio reaches 10^{42} :

$$\frac{F_{\text{em}}}{F_g} = \frac{e^2}{4\pi\epsilon_0 G m_e^2} = \alpha \left(\frac{M_{\text{Pl}}}{m_e} \right)^2 \sim 4.17 \times 10^{42}. \quad (33)$$

This immense disparity motivated Paul Dirac’s Large Numbers Hypothesis [Dirac, 1937].

RefG interprets this hierarchy through the **differing ontological character** of the two interactions:

1. **Electromagnetism** is a primary, direct transverse-phase gauge coupling between topological charges;
2. **Gravity:** An oscillon’s internal oscillatory energy induces a **secondary, Bernoulli-type static pressure deficit** in the surrounding foundation. Its long-range refractive trace is read as the gravitational field ($-\Delta P \propto \rho_{\text{dyn}}$).
 - The Planck mass ($M_{\text{Pl}} = \sqrt{\hbar c/G} \approx 2.18 \times 10^{-8} \text{ kg} \approx 1.22 \times 10^{19} \text{ GeV}$) is the limiting calibration scale for the elastic response of the foundation-medium;
 - Electrons and protons are localized oscillon modes whose energies lie many orders of magnitude below that scale ($m_e/M_{\text{Pl}} \sim 10^{-22}$).

Because the gravitational pressure deficit is proportional to oscillon energy (m_e), and gravitational attraction couples two such deficits (Gm_e^2), gravitational interaction between elementary particles is quadratically suppressed relative to the Planck scale ($m_e^2/M_{\text{Pl}}^2 \sim 10^{-44}$).

In RefG, the extreme weakness of gravity is read as a difference between these two channels: gravity is a pressure-mediated refractive effect generated by microscopic oscillons. The ratio $\alpha(M_{\text{Pl}}/m)^2$ reproduces the correct scale; its full origin requires a common dynamics for G , M_{Pl} , and the microscopic energy of an oscillon.

6 Epilogue: Boundaries of Knowledge

6.1 Why One Foundation and One Manifested Medium?

At the conclusion of this work, we return to the central physical and philosophical question: **Why a single foundation-medium?**

One primary motivation is ontological parsimony (Occam’s razor): a single substrate radically minimizes the number of independent, disconnected axiomatic inputs. However, mathematical simplicity alone is insufficient to establish physical truth.

A far deeper and more decisive argument is the **common origin of physical law**. When electrons, quarks, and photons are understood as distinct resonant and topological modes of a single universal foundation-medium, RefG seeks to derive the conservation of energy, momentum, angular momentum, and electric charge from the dynamic equilibrium of this common substrate rather than postulating them as external, unconnected axioms. Physical interactions represent couplings between differing modes of the same medium, while conservation laws express the geometric continuity of its total energy balance.

This perspective addresses Eugene Wigner’s famous reflection on the “unreasonable effectiveness of mathematics” in the natural sciences [Wigner, 1960]. The postulate of a single ontological substrate provides this profound harmony with a physical explanation: **The universal mathematical order of physical law reflects the internal dynamical coherence of a single primary substance.**

Postulating multiple mutually independent fundamental fields (as in the Standard Model) merely defers the question: *What overarching symmetry principle constrains these disparate fields to obey the same universal conservation laws and relativistic spacetime symmetries?* Within the RefG ontology, this universality is not an external decree—it is the intrinsic dynamics of the shared substrate.

This argument gives the Standard Model’s **fine-tuning problem** a common structural target:

- The stringent observational bound on the electrical neutrality of bulk matter ($|q_p + q_e|/e < 10^{-21}$);
- The observational compatibility of the photon rest mass with zero ($m_\gamma < 10^{-18}$ eV);
- The precise quantization of fractional quark charges [Navas et al., 2024].

These interrelated constraints call for a common structural explanation, which RefG associates with the topological geometry of the foundation-medium.

A single universal foundation gives rise to immense phenomenological richness: it concurrently supports volumetric pressure, shear stress, phase, topology, vortical circulation, memory, and resonance. This reflects Philip Anderson’s celebrated insight—“**More is Different**”: every emergent level of structural complexity exhibits its own autonomous physical laws [Anderson, 1972]. RefG understands transitions across scales as the self-organization of a multi-channel medium.

RefG reads the measurable cosmological Universe as a **self-measuring global resonant network**. The foundation itself is pre-spatial and is not assigned a spatial size. A mathematical witness exists in the form of a finite, connected periodic network with no unique center, but this witness does not determine the actual dimension or topology of the Universe. A finite, boundaryless global resonator remains a physical candidate. The Universe measures itself through its own resonant modes.

6.2 Three Operational Boundaries

Measurement of the Universe is always performed by components of the Universe itself: our clocks, measuring rods, detectors, and bodies are oscillon structures of the universal foundation-medium. This concept of a **self-measuring Universe** resonates with John Wheeler’s *It from Bit* program [Wheeler, 1990]. In Wheeler’s formulation, information is primary; in RefG, the physical foundation-medium is primary, while the informational trace is a result of its self-distinction.

Here, *boundary* denotes the domain within which a measuring procedure is physically defined. An observer internal to the Universe can measure only those relations that are defined within the active nodal network itself.

Kurt Gödel’s incompleteness theorems [Gödel, 1931] provide a fundamental epistemological analogy: just as a consistent formal system contains true propositions that cannot be proved from within that system, a self-measuring Universe raises the question: *How completely can an internal component of the system measure the global boundaries of the entire system?*

This self-measuring picture gives rise to three operational boundaries:

1. **The Operational Origin of Time:** In the working scenario, time begins with the first foundation-wide self-distinction, and this boundary is assigned $t = 0$ and $\tau = 0$. No physical past exists before the birth of the Universe, because change and the ordering of events are not yet defined. The history measured by present-day clocks spans approximately 13.8 Gyr. On the selected branch, the process age is also finite, while its exact value remains to be derived.
2. **The Operational Boundary of Space:** We measure the connections among distinguishable oscillon nodes. An exterior of the Universe is not a possible location of observation. The cosmological horizon is a measurement boundary defined from within the network. The quantity R_{act} is a local activation or correlation scale and does not denote an external radius of the Universe.
3. **The Operational Boundary of Scale:** In RefG, the Planck length, $\ell_{\text{Pl}} \sim 10^{-35}$ m, is the natural transition scale constructed from c , $\hbar$, and G , at which quantum and gravitational effects become simultaneously important. The exact relation between the minimum operational distinguishability threshold ℓ_* and the Planck length must be determined by the microdynamics of the foundation.

In all three cases, the boundary arises from the conditions of measurement internal to the Universe. A finite object may extend to infinity in a chosen coordinate representation, as in the stereographic

projection of a sphere onto an infinite plane or in Benoit Mandelbrot’s coastline paradox [Mandelbrot, 1982].

6.3 Methodological Principles

The methodological foundation of this research program is absolute: **An intuitive physical picture must be translated into a rigorous, computable, mathematically self-consistent, and falsifiable formalism.**

The theoretical analysis of a particle oscillon follows a rigorous 9-step verification chain:

1. **Empirical Grounding:** Precise determination of mass, charge, spin, lifetime, and scattering form factors;
2. **Mode Identification:** Selecting the appropriate resonant and topological class within the foundation-medium;
3. **Geometric Ansatz:** Formulating the spatial geometry, topological invariants (Lk, Wr, Tw), and boundary conditions;
4. **Dynamical Law:** Deriving the nonlinear field action, Lagrangian density, and equations of motion;
5. **Symmetries and Conservation Laws:** Identifying Noether currents, charge, spin, color, and zero modes;
6. **Existence and Stability:** Proving the existence of localized finite-energy soliton solutions and analyzing their perturbation spectrum;
7. **Parameter Closure:** Fixing all free coefficients independently of the data to be predicted;
8. **Novel Predictions:** Computing independent, measurable physical observables and their uncertainties;
9. **Empirical Confrontation:** Testing predictions against observational data under pre-defined falsification criteria.

A working hypothesis and a verified computational result are strictly distinguished. Every calculation is tied to reproducible code, explicit datasets, and executable verification gates.

6.4 Where This Program Could Die (The Falsification Map)

The hallmark of a legitimate scientific theory is its vulnerability to empirical refutation (Popper’s criterion). RefG establishes a transparent, three-tiered falsification map:

1. The Entire Framework is Falsified if:

1. The oscillon sector fails to recover effective Lorentz invariance and spatial isotropy of light speed ($|\Delta c/c| > 10^{-17}$; Herrmann et al. 2009), or gravitational-wave propagation speed deviates from c ($|v_{\text{gw}} - c|/c > 10^{-15}$; Abbott et al. 2017).
2. The universality of free fall between inertial and gravitational mass exhibits a composition-dependent violation exceeding observational bounds ($\eta > 10^{-15}$; MICROSCOPE: Touboul et al. 2022).
3. The global pressure evolution of the foundation-medium fails to yield a unique relation between process and metric time, or the cosmology built from that relation fails to recover the observed CMB, BBN, photon redshift, and Type Ia supernova $(1+z)$ time-dilation relations [Blondin et al., 2008].

4. Dimensionless coupling constants vary under local gravitational potential shifts ($|\Delta\alpha/\alpha| > 10^{-6}$; Kotuš et al. 2017).
5. Self-distinction of the foundation fails to establish a unified bridge to mutually distinguishable modes, local causality, and additive accounting of energy, momentum, and charge.

2. An Individual Sector is Falsified if:

1. Gravitational-wave energy flux fails to reproduce binary pulsar orbital decay to a precision of 10^{-3} – 10^{-4} [Kramer et al., 2021], or scalar breathing modes exceed LIGO-Virgo-KAGRA bounds.
2. The galactic vortex framework fails to reproduce SPARC/RAR data [McGaugh et al., 2016] or the 8σ separation between lensing centroids and X-ray gas in the Bullet Cluster [Clowe et al., 2006].
3. Combined vortical-nodal sources fail to recover primary CMB anisotropies, the BAO $100 h^{-1}$ Mpc ruler, and large-scale structure statistics [Eisenstein et al., 2005].
4. The compact regular-core model fails to match Event Horizon Telescope shadow diameters, the compact-branch oscillation spectrum, or gravitational-wave echo limits [Cardoso and Pani, 2019].

3. A Specific Geometric Reading is Falsified if:

1. The electron oscillon profile fails to maintain a pointlike electromagnetic form factor down to 10^{-18} m [Navas et al., 2024].
2. The Koide/ C_3 branch fails to construct an action-derived radiative bridge to pole masses, or a fourth sequential charged lepton is discovered.
3. Galactic-flow circulation is not quantized in units of h/m within an independently fixed effective-mass and coherence window.
4. The Möbius double-cover fails to yield the Dirac value $g = 2$, Fock-space antisymmetry, or QED radiative corrections.

Each falsification criterion is tied to specific experimental measurements: an adverse result eliminates the corresponding hypothesis from the research program rather than prompting ad-hoc verbal revisions.

Conclusion

This monograph has established the unified conceptual architecture of **Refractive Gravity** (RefG):

1. **Ontological Starting Point:** In the working scenario, one universal pre-spatial foundation undergoes a centerless initial transition that produces the first distinguishable relations. Resonant fields of oscillon seeds localized within this initial relational layer overlap; the resulting coupling increases network connectivity, while stable matter forms through later locking. The physical action governing the transition, the coupling threshold, and the law selecting stable modes remain open.
2. **Matter as Oscillons:** Locally confined, topologically stable nonlinear wave configurations of the foundation—**oscillons**—induce a static pressure deficit in the surrounding substrate via Bernoulli-type dynamics ($-\Delta P \propto \rho_{\text{dyn}}$), registered by external observers as gravitational mass.
3. **Gravity as Refraction:** The propagation of matter and light wavepackets through an inhomogeneous pressure landscape constitutes universal optical-geometric refraction; along the static weak-field 1PN branch, this mechanism recovers the predictions of General Relativity (PPN $\beta = \gamma = 1$).

This unified pressure chain unfolds across three primary astrophysical scales:

- **Locally:** In compact relativistic objects, where internal topological content and externally read mass decouple via two-channel accounting;
- **Galactically:** In the coherent vortical circulation and polarization of the foundation, which RefG connects to MOND and Tully–Fisher (RAR/BTFR) regularities without dark matter particles;
- **Cosmologically:** The framework combines a candidate Genesis scenario beginning at a global, centerless boundary with a selected expansion–relaxation branch. Conditional on this branch, decreasing background pressure reduces operational material size and externally read mass, while growth of the foundation scale a and contraction of the material standard p combine in the observable scale $A = a/p$. The histories $a(t)$, $P_F(t)$, and $p(t)$ remain to be derived. Type Ia supernova time dilation, the process maturity of early galaxies, and the pressure-based reading of the cosmological constant are separate tests of this cosmological program.

Concurrently, the framework connects material discreteness to the candidate eigenspectrum of a three-dimensional volumetric resonator and to population-locked synchronization; C_3 generation symmetry and the ninth-order cycle ($\theta = 2/9$, $A_K = \sqrt{2}$) provide a geometric map of Koide’s lepton mass formula; and the topological invariants $Lk = Wr + Tw$ organize spin, charge, and color quantum numbers within one geometric language.

Refractive Gravity constructs a comprehensive research program aimed at deriving a unified, computable, and empirically testable account of gravitation, matter, and measurability from a single ontological foundation. This monograph establishes a firm foundation for that physical paradigm and opens a broad path for further theoretical and experimental research.

Current Status of the Research Program

The current status of each theoretical component is cataloged here to maintain clarity and transparency across the research program:

1. **Ontological Foundation, Cosmic Genesis, and the Origin of Measurability**—The selected working scenario combines a centerless initial boundary common to the entire foundation, the first minimally distinguishable relations, localization of oscillon seeds, overlap of resonant fields, coupling-mediated network connection, and subsequent stabilization. W3-37 separates the stipulated parameter domains and verifies that the selected event ordering contains no closed causal cycle; W3-38 gives a finite periodic-graph witness that connectivity and the absence of a unique center are compatible; and W3-39 verifies a conditional post-origin ledger of energy-transfer signs in predeclared sectors. In a finite symmetric-network example from the same W3-39 report, where all vertices are equivalent, a record strictly ordered by distance from one selected center violates the stipulated centerlessness. These reports do not derive a physical Genesis mechanism. The transition trigger and action, actual dimension and topology, an oscillon-seed solution, the resonant-field equation, the physical coupling threshold, the initial energy distribution, the thermal history, and observational consequences remain open.
2. **Inter-Node Separation and Minimum Distinguishability**—The operational concept of distinguishability has been established; a Planck-order threshold is a physical candidate, but it has not been derived from the underlying action.
3. **Local Collective Cadence**—Common-rescaling invariance and trial models of mutual synchronization have been obtained; calculating the complete spectrum of stable oscillons remains an active task.

4. **Oscillon Deficit, Scale, Time, and Mass**—Metric filters and the first-order biconformal branch have been derived symbolically; a finite-energy nonlinear oscillon profile and the full mass-readout map remain open.
5. **Weak-Field Gravity and EFT Architecture**—On the pressure-deficit channel, $H = 0$, $\gamma = 1$, and $\beta = 1$ are obtained. From conservation of oscillon topological-defect density, macroscopic four-dimensional covariance, and the effective postulates of Fierz–Pauli dynamics, Lovelock’s and Deser’s theorems yield a conditional low-energy General-Relativistic architecture within effective field theory. A direct nonlinear derivation from the microdynamics remains open.
6. **Inertia and Gravitational Mass**—The Noether identity is conditionally established for finite-energy, Lorentz-covariant localized solutions satisfying the Laue boundary condition; an exact derivation of the zero mode remains open.
7. **Gravitational Waves and Local Unobservability**—In the local Minkowski limit, tensor propagation speed, two transverse-traceless (TT) polarizations arising from shear modes, and detector response are established; cosmological propagation, source-flux normalization, the full waveform, and scalar suppression bounds remain open.
8. **Compact Objects**—A static, conditional map and finite-core matching of the C^2 ansatz have been obtained; within one specific exterior-layer and pressure-reversal criterion, admissible domains do not overlap. The physical equation of state, gravitational collapse, rotation, and global stability remain open.
9. **Galaxy Rotation**—An empirical test using $B - V$ colors did not detect the anticipated age correlation. This result is compatible with common rescaling, but it does not establish either the Doppler-cancellation mechanism or a complete physical explanation of the MOND correlations; both remain open (W3-18). The symbolic algebraic cancellation $c_{\text{coord}}/c_0 = p^2$ has been obtained, but direct physical confirmation from PHANGS/SPARC data and an exact process-level derivation of vortex rotation are still lacking.
10. **Cosmological Relaxation and the Time Bridge**— $A = a/p$ is the exact definition of the operational scale. The selected cosmological branch permits both expansion of the foundation scale a and contraction of the material standard p , but neither history has yet been derived. The approximately 13.8 Gyr age calibrated by present-day clocks is an observational cosmological quantity; finiteness of process history is an initial condition of this branch, and its exact duration requires dynamics for $P_F(t)$. A full derivation of Type Ia supernova $(1 + z)$ time dilation requires a map of photon propagation and atomic standards, while interpreting early galaxies requires an independent model of structural growth. Both channels are ongoing tests of the program.
11. **Two-Level Time and Quantum Randomness**—The interpretive framework is compatible with Bell tests and delayed-choice experiments; formal derivations of Born’s rule, the no-signaling condition, and measurement dynamics remain open.
12. **Particles as Oscillon Modes**—The geometric placement map and initial spectral classification have been formulated as candidates; finite-energy oscillons, form factors, masses, couplings, and decay channels remain open.
13. **The Koide/ $C_3/2/9$ Structure**—The C_3 algebra yields $K = 2/3$ exactly; with $(A_K, \theta) = (\sqrt{2}, 2/9)$ anchored to the electron pole mass, tree-level predictions deviate by 451.7σ for the muon and 0.61σ for the tau. The choice $h = 2$, $m \propto \nu^2$, action-level derivation, and radiative corrections remain open.
14. **Spin, Pauli Exclusion, and Fermionic Statistics**—The Möbius double-covered geometry provides a candidate mechanism; a complete derivation of Fock space and the Spin-Statistics Theorem from first principles remains open.

15. **Charge, the Photon, and the Electromagnetic Channel**—The following conditional structural results have been obtained: local-frame $U(1)$, two helicities, screening of the hedgehog’s otherwise unscreened asymptotic sigma-field component, whose energy grows linearly with radius, together with the surviving $1/r^2$ far field and flux accounting after screening, and subtheorems for a conserved phase current. Canonical charge normalization, Maxwell’s equations, Gauss’s law, Ward identities, quantum electrodynamics, the Dirac sector, and the exact pressure-deficit gravitational source channel of freely propagating radiation remain open.
16. **Color, Quarks, and Gluons**—The three-color, eight-mode construction provides only a partially compatible structural map; $SU(3)$ structure constants, interaction vertices, scale evolution, the confinement potential, and fractional charge normalization remain open.
17. **Electroweak Map and the Higgs**—This constitutes a geometric placement map; $SU(2)_L$ chirality, a W/Z mass matrix preserving a massless photon, CKM/PMNS mixing, neutrino mass hierarchy, Yukawa couplings, and the vacuum expectation value remain open.
18. **The Fine-Structure Constant α** —A nodal interpretation and target-independent spectral measure have been obtained: α is read from the relative strengths of the electric-flux and transverse-photon response channels, with the overall scale canceling. A microscopic action for physical charge Q , a unique numerical value, the complete charged spectrum, and RG flow to the Thomson limit remain open. The earlier boundary formula remains a target-dependent conditional result.
19. **The Gravitational Hierarchy**—The force ratio is expressed algebraically as $F_{\text{em}}/F_g = \alpha(M_{\text{Pl}}/m)^2$; the fundamental origins of G , M_{Pl} , the oscillon energy scale, and the mass hierarchy remain open.
20. **A Self-Measuring Manifested Universe**—The global-resonator picture is a cosmological-level hypothesis. The finite periodic graph of W3-38 shows only that connectivity and the absence of a unique graph-theoretic center can coexist in one mathematical construction; it does not derive cosmic finiteness, boundarylessness, physical dimension, or topology. Low-multipole and topological signatures remain candidate observables.

References

- B. P. Abbott et al. Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116:061102, 2016. LIGO Scientific Collaboration and Virgo Collaboration.
- B. P. Abbott et al. Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A. *The Astrophysical Journal Letters*, 848:L13, 2017. LIGO Scientific Collaboration, Virgo Collaboration, Fermi-GBM, and INTEGRAL.
- P. W. Anderson. More is different. *Science*, 177(4047):393–396, 1972.
- R. Arnowitt, S. Deser, and C. W. Misner. The dynamics of general relativity. In L. Witten, editor, *Gravitation: An Introduction to Current Research*, pages 227–265. Wiley, New York, 1962.
- C. Barceló, S. Liberati, and M. Visser. Analogue gravity. *Living Reviews in Relativity*, 14:3, 2011. First version: *Living Reviews in Relativity* 8, 12 (2005).
- J. Barckicke, E. Falcon, and C. Gissinger. Kelvin wave propagation along vortex cores. *Nature Physics*, 22:409–414, 2026.
- J. D. Bekenstein. Relativistic gravitation theory for the modified newtonian dynamics paradigm. *Physical Review D*, 70:083509, 2004.

- J. S. Bell. On the Einstein–Podolsky–Rosen paradox. *Physics Physique Fizika*, 1:195–200, 1964.
- A. Berera, R. V. Buniy, T. W. Kephart, H. Päs, and J. G. Rosa. Knotty inflation and the dimensionality of spacetime. *European Physical Journal C*, 77:611, 2017.
- L. Berezhiani and J. Khoury. Theory of dark matter superfluidity. *Physical Review D*, 92:103510, 2015.
- D. Bernoulli. *Hydrodynamica, sive de viribus et motibus fluidorum commentarii*. Johann Reinhold Dulsecker, Strasbourg, 1738.
- M. V. Berry. Quantal phase factors accompanying adiabatic changes. *Proceedings of the Royal Society of London A*, 392:45–57, 1984.
- B. Bertotti, L. Iess, and P. Tortora. A test of general relativity using radio links with the Cassini spacecraft. *Nature*, 425:374–376, 2003.
- V. Bjerknes. *Fields of Force*. Columbia University Press, New York, 1906.
- S. Blondin, T. M. Davis, K. Krisciunas, B. P. Schmidt, J. Sollerman, W. M. Wood-Vasey, et al. Time dilation in Type Ia supernova spectra at high redshift. *Astrophysical Journal*, 682:724–736, 2008.
- I. L. Bogolyubsky and V. G. Makhankov. On the pulsed soliton lifetime in two classical relativistic theory models. *JETP Letters*, 24:12–14, 1976.
- D. Bohm. A suggested interpretation of the quantum theory in terms of “hidden” variables. I & II. *Physical Review*, 85:166–179, 180–193, 1952.
- M. Born. Zur quantenmechanik der stoßvorgänge. *Zeitschrift für Physik*, 37:863–867, 1926.
- C. A. Brannen. The lepton masses. Unpublished preprint, 2006.
- C. A. Brannen. Spin path integrals and generations. *Foundations of Physics*, 40:1681–1699, 2010.
- P. W. Bridgman. *The Logic of Modern Physics*. Macmillan, New York, 1927.
- G. Călugăreanu. Sur les classes d’isotopie des nœuds tridimensionnels et leurs invariants. *Czechoslovak Mathematical Journal*, 11(4):588–625, 1961.
- V. Cardoso and P. Pani. Testing the nature of dark compact objects: A status report. *Living Reviews in Relativity*, 22:4, 2019.
- G. F. Chew and S. C. Frautschi. Principle of equivalence for all strongly interacting particles within the S -matrix framework. *Physical Review Letters*, 7:394–397, 1961.
- E. F. F. Chladni. *Entdeckungen über die Theorie des Klanges*. Weidmanns Erben und Reich, Leipzig, 1787.
- W. K. Clifford. On the space-theory of matter. *Proceedings of the Cambridge Philosophical Society*, 2: 157–158, 1876. Read 21 February 1870; printed in 1876.
- D. Clowe, M. Bradač, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, and D. Zaritsky. A direct empirical proof of the existence of dark matter. *Astrophysical Journal Letters*, 648:L109–L113, 2006.
- S. Coleman. Q -balls. *Nuclear Physics B*, 262(2):263–283, 1985. Addendum: Nuclear Physics B 269, 744 (1986).
- E. J. Copeland, M. Gleiser, and H.-R. Müller. Oscillons: Resonant configurations during bubble collapse. *Physical Review D*, 52:1920–1933, 1995.

- P. Delva, N. Puchades, E. Schönemann, et al. Gravitational redshift test using eccentric Galileo satellites. *Physical Review Letters*, 121:231101, 2018.
- S. Deser. Self-interaction and gauge invariance. *General Relativity and Gravitation*, 1:9–18, 1970.
- B. S. DeWitt. Quantum theory of gravity. I. the canonical theory. *Physical Review*, 160:1113–1148, 1967.
- R. H. Dicke. Gravitation without a principle of equivalence. *Reviews of Modern Physics*, 29:363–376, 1957.
- P. A. M. Dirac. The cosmological constants. *Nature*, 139:323, 1937.
- P. Ehrenfest. In what way does it become manifest in the fundamental laws of physics that space has three dimensions? *Proceedings of the Koninklijke Akademie van Wetenschappen te Amsterdam*, 20: 200–209, 1918.
- A. Einstein. Zur elektrodynamik bewegter körper. *Annalen der Physik*, 17:891–921, 1905. English title: On the Electrodynamics of Moving Bodies.
- A. Einstein. Über den einfluß der schwerkraft auf die ausbreitung des lichtes. *Annalen der Physik*, 35: 898–908, 1911. Band 340.
- A. Einstein. Lichtgeschwindigkeit und statik des gravitationsfeldes. *Annalen der Physik*, 38:355–369, 1912. Band 343.
- A. Einstein. Die grundlage der allgemeinen relativitätstheorie. *Annalen der Physik*, 354(7):769–822, 1916.
- D. J. Eisenstein et al. Detection of the baryon acoustic peak in the large-scale correlation function of SDSS luminous red galaxies. *The Astrophysical Journal*, 633:560–574, 2005.
- Event Horizon Telescope Collaboration. First M87 event horizon telescope results. I. the shadow of the supermassive black hole. *The Astrophysical Journal Letters*, 875:L1, 2019.
- L. Faddeev and A. J. Niemi. Stable knot-like structures in classical field theory. *Nature*, 387:58–61, 1997.
- X. Fan, T. G. Myers, B. A. D. Sukra, and G. Gabrielse. Measurement of the electron magnetic moment. *Physical Review Letters*, 130:071801, 2023.
- R. P. Feynman. Application of quantum mechanics to liquid helium. In C. J. Gorter, editor, *Progress in Low Temperature Physics*, volume 1, pages 17–53. North-Holland, Amsterdam, 1955.
- R. P. Feynman. *QED: The Strange Theory of Light and Matter*. Princeton University Press, Princeton, 1985.
- G. F. FitzGerald. The ether and the earth’s atmosphere. *Science*, 13(328):390, 1889.
- H. Fizeau. Sur les hypothèses relatives à l’éther lumineux, et sur une expérience qui paraît démontrer que le mouvement des corps change la vitesse avec laquelle la lumière se propage dans leur intérieur. *Comptes Rendus de l’Académie des Sciences*, 33:349–355, 1851.
- R. Foot. A note on Koide’s lepton mass relation. Unpublished preprint, 1994.
- F. B. Fuller. The writhing number of a space curve. *Proceedings of the National Academy of Sciences USA*, 68(4):815–819, 1971.
- M. Gleiser. Pseudostable bubbles. *Physical Review D*, 49:2978–2981, 1994.

- K. Gödel. Über formal unentscheidbare sätze der Principia Mathematica und verwandter systeme I. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.
- W. Gordon. Zur lichtfortpflanzung nach der relativitätstheorie. *Annalen der Physik*, 72:421–456, 1923. Band 377.
- B. Hensen, H. Bernien, A. E. Dréau, et al. Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature*, 526:682–686, 2015.
- S. Herrmann, A. Senger, K. Möhle, M. Nagel, E. V. Kovalchuk, and A. Peters. Rotating optical cavity experiment testing lorentz invariance at the 10^{-17} level. *Physical Review D*, 80:105011, 2009.
- S. Herrmann, F. Finke, M. Lülf, et al. Test of the gravitational redshift with Galileo satellites in an eccentric orbit. *Physical Review Letters*, 121:231102, 2018.
- W. Hu, R. Barkana, and A. Gruzinov. Fuzzy cold dark matter: The wave properties of ultralight particles. *Physical Review Letters*, 85:1158–1161, 2000.
- L. Hui, A. Joyce, M. J. Landry, and X. Li. Vortices and waves in light dark matter. *Journal of Cosmology and Astroparticle Physics*, 2021(01):011, 2021.
- C. Huygens. Letters of february 1665 on the “sympathy” of two pendulum clocks. In *Oeuvres complètes de Christiaan Huygens*, volume 5. Société Hollandaise des Sciences / Martinus Nijhoff, The Hague, 1665. In Correspondance 1664–1665; published in 1893.
- T. Jacobson. Thermodynamics of spacetime: The Einstein equation of state. *Physical Review Letters*, 75:1260–1263, 1995.
- V. Jacques, E. Wu, F. Grosshans, F. Treussart, P. Grangier, A. Aspect, and J.-F. Roch. Experimental realization of Wheeler’s delayed-choice gedanken experiment. *Science*, 315:966–968, 2007.
- B. D. Josephson. Possible new effects in superconductive tunnelling. *Physics Letters*, 1:251–253, 1962.
- T. W. B. Kibble. Topology of cosmic domains and strings. *Journal of Physics A: Mathematical and General*, 9:1387–1398, 1976.
- H. Kleinert. *Multivalued Fields in Condensed Matter, Electromagnetism, and Gravitation*. World Scientific, Singapore, 2008.
- F. R. Klinkhamer and G. E. Volovik. Dynamic vacuum variable and equilibrium approach in cosmology. *Physical Review D*, 78:063528, 2008.
- Y. Koide. Fermion-boson two-body model of quarks and leptons and Cabibbo mixing. *Lettere al Nuovo Cimento*, 34:201–205, 1982.
- Y. Koide. New view of quark and lepton mass hierarchy. *Physical Review D*, 28:252–254, 1983. Erratum: *Physical Review D* 29, 1544 (1984).
- A. Komar. Covariant conservation laws in general relativity. *Physical Review*, 113:934–936, 1959.
- S. M. Kotuš, M. T. Murphy, and R. F. Carswell. High-precision limit on variation in the fine-structure constant from a single quasar absorption system. *Monthly Notices of the Royal Astronomical Society*, 464:3679–3703, 2017.
- M. Kramer, I. H. Stairs, R. N. Manchester, et al. Strong-field gravity tests with the double pulsar. *Physical Review X*, 11:041050, 2021.

- Y. Kuramoto. Self-entrainment of a population of coupled non-linear oscillators. In H. Araki, editor, *International Symposium on Mathematical Problems in Theoretical Physics*, volume 39 of *Lecture Notes in Physics*, pages 420–422. Springer, Berlin, 1975.
- I. Labbé, P. van Dokkum, E. Nelson, R. Bezanson, K. A. Suess, J. Leja, et al. A population of red candidate massive galaxies ~ 600 Myr after the Big Bang. *Nature*, 616:266–269, 2023.
- F. Lelli, S. S. McGaugh, and J. M. Schombert. SPARC: Mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves. *The Astronomical Journal*, 152:157, 2016.
- M. A. Levin and X.-G. Wen. String-net condensation: A physical mechanism for topological phases. *Physical Review B*, 71:045110, 2005.
- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration. Tests of general relativity with GWTC-3. *Physical Review D*, 105:082002, 2022.
- H. A. Lorentz. De relatieve beweging van de aarde en den aether. *Verslagen der Zittingen van de Wis- en Natuurkundige Afdeeling, Koninklijke Akademie van Wetenschappen te Amsterdam*, 1:74–79, 1892.
- G. Lotishvili. RefG: Refractive gravity verification gates (SymPy). Zenodo, 2026a.
- G. Lotishvili. Quantum extensions of bi-conformal scalar-tensor gravity: Particles as spacetime resonances, the Bernoulli mechanism for gravity, and falsifiable predictions. Zenodo, 2026b.
- D. Lovelock. The einstein tensor and its generalizations. *Journal of Mathematical Physics*, 12:498–501, 1971.
- E. Mach. *Die Mechanik in ihrer Entwicklung, historisch-kritisch dargestellt*. F. A. Brockhaus, Leipzig, 1883. English translation: *The Science of Mechanics*, Open Court, Chicago (1893).
- B. B. Mandelbrot. *The Fractal Geometry of Nature*. W. H. Freeman and Company, San Francisco, 1982.
- P. O. Mazur and E. Mottola. Gravitational vacuum condensate stars. *Proceedings of the National Academy of Sciences USA*, 101:9545–9550, 2004. First circulated as “Gravitational Condensate Stars: An Alternative to Black Holes,” arXiv:gr-qc/0109035 (2001).
- S. S. McGaugh, F. Lelli, and J. M. Schombert. Radial acceleration relation in rotationally supported galaxies. *Physical Review Letters*, 117:201101, 2016.
- J. Michell. On the means of discovering the distance, magnitude, &c. of the fixed stars, in consequence of the diminution of the velocity of their light... *Philosophical Transactions of the Royal Society of London*, 74:35–57, 1784.
- A. A. Michelson and E. W. Morley. On the relative motion of the earth and the luminiferous ether. *American Journal of Science*, 34(203):333–345, 1887. Series 3.
- M. Milgrom. A modification of the newtonian dynamics as a possible alternative to the hidden mass hypothesis. *Astrophysical Journal*, 270:365–370, 1983.
- Y. Minowa, Y. Yasui, T. Nakagawa, S. Inui, M. Tsubota, and M. Ashida. Direct excitation of Kelvin waves on quantized vortices. *Nature Physics*, 21:233–238, 2025.
- H. K. Moffatt. The degree of knottedness of tangled vortex lines. *Journal of Fluid Mechanics*, 35: 117–129, 1969.
- J. R. Muñoz de Nova, K. Golubkov, V. I. Kolobov, and J. Steinhauer. Observation of thermal Hawking radiation and its temperature in an analogue black hole. *Nature*, 569:688–691, 2019.

- S. Navas et al. Review of particle physics. *Physical Review D*, 110:030001, 2024. Particle Data Group.
- E. Noether. Invariante variationsprobleme. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1918:235–257, 1918.
- L. Onsager. Statistical hydrodynamics. *Il Nuovo Cimento (Supplemento)*, 6:279–287, 1949.
- D. N. Page and W. K. Wootters. Evolution without evolution: Dynamics described by stationary observables. *Physical Review D*, 27:2885, 1983.
- W. Pauli. The connection between spin and statistics. *Physical Review*, 58:716–722, 1940.
- R. Penrose. Gravitational collapse and space-time singularities. *Physical Review Letters*, 14:57–59, 1965.
- M. Planck. Über irreversible strahlungsvorgänge. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin*, 5:440–480, 1899.
- Planck Collaboration. Planck 2015 results. XVIII. background geometry and topology of the universe. *Astronomy & Astrophysics*, 594:A18, 2016. arXiv:1502.01593.
- Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.
- R. V. Pound and G. A. Rebka, Jr. Apparent weight of photons. *Physical Review Letters*, 4:337–341, 1960.
- H. E. Puthoff. Polarizable-vacuum (PV) approach to general relativity. *Foundations of Physics*, 32: 927–943, 2002.
- A. F. Rañada. A topological theory of the electromagnetic field. *Letters in Mathematical Physics*, 18: 97–106, 1989.
- R. L. Ricca and B. Nipoti. Gauss’ linking number revisited. *Journal of Knot Theory and Its Ramifications*, 20(10):1325–1343, 2011.
- V. C. Rubin and W. K. Ford, Jr. Rotation of the Andromeda nebula from a spectroscopic survey of emission regions. *Astrophysical Journal*, 159:379–403, 1970.
- A. D. Sakharov. Vacuum quantum fluctuations in curved space and the theory of gravitation. *Doklady Akademii Nauk SSSR*, 177:70–71, 1967. English translation: Soviet Physics Doklady 12, 1040–1041 (1968); reprinted in General Relativity and Gravitation 32, 365–367 (2000).
- J. Schwinger. On quantum-electrodynamics and the magnetic moment of the electron. *Physical Review*, 73:416–417, 1948.
- I. I. Shapiro. Fourth test of general relativity. *Physical Review Letters*, 13:789–791, 1964.
- S. Shapiro. Josephson currents in superconducting tunneling: The effect of microwaves and other observations. *Physical Review Letters*, 11:80–82, 1963.
- K. Shulga. Charged-lepton koide geometry from a green-dressed compact family cycle, 2026.
- C. Skordis and T. Złośnik. New relativistic theory for modified newtonian dynamics. *Physical Review Letters*, 127:161302, 2021.
- T. H. R. Skyrme. A non-linear field theory. *Proceedings of the Royal Society of London A*, 260:127–138, 1961.

- Y. Sumino. Family gauge symmetry and Koide's mass formula. *Physics Letters B*, 671(4–5):477–480, 2009.
- P. G. Tait. On knots. *Transactions of the Royal Society of Edinburgh*, 28(1):145–190, 1877.
- W. Thomson. On vortex atoms. *Proceedings of the Royal Society of Edinburgh*, 6:94–105, 1867. Lord Kelvin; reprinted in *Philosophical Magazine* 34, 15–24 (1867).
- W. Thomson. On vortex motion. *Transactions of the Royal Society of Edinburgh*, 25:217–260, 1869. Lord Kelvin.
- A. Tonomura, J. Endo, T. Matsuda, T. Kawasaki, and H. Ezawa. Demonstration of single-electron buildup of an interference pattern. *American Journal of Physics*, 57(2):117–120, 1989.
- P. Touboul, G. Métris, M. Rodrigues, et al. MICROSCOPE mission: Final results of the test of the equivalence principle. *Physical Review Letters*, 129:121102, 2022.
- R. B. Tully and J. R. Fisher. A new method of determining distances to galaxies. *Astronomy and Astrophysics*, 54:661, 1977.
- W. G. Unruh. Experimental black-hole evaporation? *Physical Review Letters*, 46:1351–1353, 1981.
- E. Verlinde. On the origin of gravity and the laws of newton. *Journal of High Energy Physics*, 2011:29, 2011.
- R. F. C. Vessot, M. W. Levine, E. M. Mattison, et al. Test of relativistic gravitation with a space-borne hydrogen maser. *Physical Review Letters*, 45:2081–2084, 1980.
- W. F. Vinen. The detection of single quanta of circulation in liquid helium II. *Proceedings of the Royal Society of London A*, 260:218–236, 1961.
- G. E. Volovik. *The Universe in a Helium Droplet*, volume 117 of *International Series of Monographs on Physics*. Clarendon Press / Oxford University Press, Oxford, 2003.
- M. von Laue. Zur dynamik der relativitätstheorie. *Annalen der Physik*, 340:524–542, 1911.
- S. Weinberg and E. Witten. Limits on massless particles. *Physics Letters B*, 96(1–2):59–62, 1980.
- J. M. Weisberg and Y. Huang. Relativistic measurements from timing the binary pulsar PSR B1913+16. *The Astrophysical Journal*, 829:55, 2016.
- C. Wetterich. A universe without expansion. *Physics of the Dark Universe*, 2:184–187, 2013.
- J. A. Wheeler. Geons. *Physical Review*, 97(2):511–536, 1955.
- J. A. Wheeler. The ‘past’ and the ‘delayed-choice’ double-slit experiment. In A. R. Marlow, editor, *Mathematical Foundations of Quantum Theory*, pages 9–48. Academic Press, 1978.
- J. A. Wheeler. Information, physics, quantum: The search for links. In W. H. Zurek, editor, *Complexity, Entropy, and the Physics of Information*, pages 3–28. Addison-Wesley, Redwood City, 1990.
- J. H. White. Self-linking and the Gauss integral in higher dimensions. *American Journal of Mathematics*, 91(3):693–728, 1969.
- G. J. Whitrow. Why physical space has three dimensions. *British Journal for the Philosophy of Science*, 6:13–31, 1955.
- E. P. Wigner. The unreasonable effectiveness of mathematics in the natural sciences. *Communications on Pure and Applied Mathematics*, 13(1):1–14, 1960.

- C. M. Will. The confrontation between general relativity and experiment. *Living Reviews in Relativity*, 17:4, 2014.
- F. Winterberg. *The Planck Aether Hypothesis: An Attempt for a Finitistic Non-Archimedean Theory of Elementary Particles*. Gauss Scientific Press, Reno, 2002.
- Ya. B. Zel'dovich. Gravitational instability: An approximate theory for large density perturbations. *Astronomy & Astrophysics*, 5:84–89, 1970.
- F. Zwicky. Die rotverschiebung von extragalaktischen nebeln. *Helvetica Physica Acta*, 6:110–127, 1933.